Main Category	Category	Terms
Disease	OMIM_DISEASE	Ataxia-telangiectasia-like disorder 2
Disease	UP_KW_DISEASE	Deafness
Disease	UP_KW_DISEASE	Disease variant
Disease	UP_KW_DISEASE	Dwarfism
Disease	UP_KW_DISEASE	Neurodegeneration
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	DNA damage
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	DNA repair
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	DNA replication
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Host-virus interaction
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	DNA-binding
Functional Annotations	UP_KW_PTM	Acetylation
Functional Annotations	UP_KW_PTM	Methylation
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_KW_PTM	Ubl conjugation
Functional Annotations	UP_KW_PTM	Disulfide bond
Functional Annotations	UP_KW_PTM	Isopeptide bond
Functional Annotations	UP_SEQ_FEATURE	CROSSLNK:Glycyl lysine isopeptide (Lys-Gly) (interchain with G-Cter in SUMO2)
Functional Annotations	UP_SEQ_FEATURE	CROSSLNK:Glycyl lysine isopeptide (Lys-Gly) (interchain with G-Cter in SUMO2); alternate
Functional Annotations	UP_SEQ_FEATURE	CROSSLNK:Glycyl lysine isopeptide (Lys-Gly) (interchain with G-Cter in ubiquitin); alternate
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Proliferating cell nuclear antigen PCNA C-terminal
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Proliferating cell nuclear antigen PCNA N-terminal
Functional	UP_SEQ_FEATURE	MUTAGEN:A->G: Complete loss of interaction with

Main Category	Category	Terms
Annotations		UHRF2.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:I->A: Complete loss of interaction with UHRF2.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->A: Inhibits recruitment to DNA damage sites, but nuclear localization is similar as the wild-type; in association with A-77.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->A: Inhibits recruitment to DNA damage sites, but nuclear localization is similar as the wild-type; in association with A-80.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Abolishes ubiquitination. No effect on interaction with SHPRH.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Inhibits acetylation, recruitment to DNA damage sites, inducible ubiquitination and protein degradation, DNA replication and repair synthesis efficiencies, but homotrimer formation, nuclear recruitment to DNA damage sites, interactions with CREBBP, EP300 and POLD1 are similar as the wild-type; in association with R-13; R-14; R-20 and R-77.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Inhibits acetylation, recruitment to DNA damage sites, inducible ubiquitination and protein degradation, DNA replication and repair synthesis efficiencies, but homotrimer formation, nuclear recruitment to DNA damage sites, interactions with CREBBP, EP300 and POLD1 are similar as the wild-type; in association with R-13; R-14; R-20 and R-80.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Inhibits acetylation, recruitment to DNA damage sites, inducible ubiquitination and protein degradation, DNA replication and repair synthesis efficiencies, but homotrimer formation, nuclear recruitment to DNA damage sites, interactions with CREBBP, EP300 and POLD1 are similar as the wild-type; in association with R-13; R-14; R-77 and R-80.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Inhibits acetylation, recruitment to DNA damage sites, inducible ubiquitination and protein degradation, DNA replication and repair synthesis efficiencies, but homotrimer formation, nuclear recruitment to DNA damage sites, interactions with CREBBP, EP300 and POLD1 are similar as the wild-type; in association with R-13; R-20; R-77 and R-80.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Inhibits acetylation, recruitment to DNA damage sites, inducible ubiquitination and protein degradation, DNA replication and repair synthesis efficiencies, but homotrimer formation, nuclear recruitment to DNA damage sites, interactions with CREBBP, EP300 and POLD1 are similar as the wild-type; in association with R-14; R-20; R-77 and R-80.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:LAPK->AAAA: Decrease in POLD3-binding.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:M->A: Complete loss of interaction with UHRF2.

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:QLGI->AAAA: Strong decrease in POLD3-binding. Impairs binding to ALKBH2.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:SHV->AAA: No effect on POLD3-binding. Impairs binding to ALKBH2.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:VDK->AAA: No effect on POLD3-binding. No effect on ALKBH2-binding.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->A: Complete loss of interaction with UHRF2.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->F: Alters chromatin-associated PCNA stability and its function in DNA replication and repair.
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with NUDT15
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	DNA replication
Gene Ontology	GOTERM_BP_DIRECT	leading strand elongation
Gene Ontology	GOTERM_BP_DIRECT	regulation of DNA replication
Gene Ontology	GOTERM_BP_DIRECT	DNA repair
Gene Ontology	GOTERM_BP_DIRECT	base-excision repair, gap-filling
Gene Ontology	GOTERM_BP_DIRECT	mismatch repair
Gene Ontology	GOTERM_BP_DIRECT	chromatin organization
Gene Ontology	GOTERM_BP_DIRECT	DNA damage response
Gene Ontology	GOTERM_BP_DIRECT	response to oxidative stress
Gene Ontology	GOTERM_BP_DIRECT	heart development
Gene Ontology	GOTERM_BP_DIRECT	translesion synthesis
Gene Ontology	GOTERM_BP_DIRECT	epithelial cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	replication fork processing

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of deoxyribonuclease activity
Gene Ontology	GOTERM_BP_DIRECT	response to estradiol
Gene Ontology	GOTERM_BP_DIRECT	response to lipid
Gene Ontology	GOTERM_BP_DIRECT	cellular response to UV
Gene Ontology	GOTERM_BP_DIRECT	estrous cycle
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of DNA repair
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of DNA replication
Gene Ontology	GOTERM_BP_DIRECT	response to cadmium ion
Gene Ontology	GOTERM_BP_DIRECT	cellular response to hydrogen peroxide
Gene Ontology	GOTERM_BP_DIRECT	cellular response to xenobiotic stimulus
Gene Ontology	GOTERM_BP_DIRECT	response to dexamethasone
Gene Ontology	GOTERM_BP_DIRECT	liver regeneration
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of DNA-directed DNA polymerase activity
Gene Ontology	GOTERM_BP_DIRECT	response to L-glutamate
Gene Ontology	GOTERM_BP_DIRECT	mitotic telomere maintenance via semi-conservative replication
Gene Ontology	GOTERM_CC_DIRECT	cyclin-dependent protein kinase holoenzyme complex
Gene Ontology	GOTERM_CC_DIRECT	chromosome, telomeric region
Gene Ontology	GOTERM_CC_DIRECT	chromatin
Gene Ontology	GOTERM_CC_DIRECT	male germ cell nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	nuclear lamina

Main Category	Category	Terms
Gene Ontology	GOTERM_CC_DIRECT	nucleoplasm
Gene Ontology	GOTERM_CC_DIRECT	replication fork
Gene Ontology	GOTERM_CC_DIRECT	centrosome
Gene Ontology	GOTERM_CC_DIRECT	nuclear body
Gene Ontology	GOTERM_CC_DIRECT	replisome
Gene Ontology	GOTERM_CC_DIRECT	nuclear replication fork
Gene Ontology	GOTERM_CC_DIRECT	PCNA complex
Gene Ontology	GOTERM_CC_DIRECT	extracellular exosome
Gene Ontology	GOTERM_CC_DIRECT	PCNA-p21 complex
Gene Ontology	GOTERM_MF_DIRECT	purine-specific mismatch base pair DNA N-glycosylase activity
Gene Ontology	GOTERM_MF_DIRECT	DNA binding
Gene Ontology	GOTERM_MF_DIRECT	chromatin binding
Gene Ontology	GOTERM_MF_DIRECT	damaged DNA binding
Gene Ontology	GOTERM_MF_DIRECT	protein binding
Gene Ontology	GOTERM_MF_DIRECT	enzyme binding
Gene Ontology	GOTERM_MF_DIRECT	nuclear estrogen receptor binding
Gene Ontology	GOTERM_MF_DIRECT	DNA polymerase processivity factor activity
Gene Ontology	GOTERM_MF_DIRECT	receptor tyrosine kinase binding
Gene Ontology	GOTERM_MF_DIRECT	dinucleotide insertion or deletion binding
Gene Ontology	GOTERM_MF_DIRECT	MutLalpha complex binding
Gene Ontology	GOTERM_MF_DIRECT	histone acetyltransferase binding

Main Category	Category	Terms
Gene Ontology	GOTERM_MF_DIRECT	identical protein binding
Gene Ontology	GOTERM_MF_DIRECT	protein-containing complex binding
Gene Ontology	GOTERM_MF_DIRECT	DNA polymerase binding
General Annotations	CHROMOSOME	20
General Annotations	CYTOGENETIC_LOCATION	20p12.3
General Annotations	ENTREZ_GENE_SUMMARY	The protein encoded by this gene is found in the nucleus and is a cofactor of DNA polymerase delta. The encoded protein acts as a homotrimer and helps increase the processivity of leading strand synthesis during DNA replication. In response to DNA damage, this protein is ubiquitinated and is involved in the RAD6-dependent DNA repair pathway. Two transcript variants encoding the same protein have been found for this gene. Pseudogenes of this gene have been described on chromosome 4 and on the X chromosome. [provided by RefSeq, Jul 2008]
General Annotations	UP_COMMENT_DISEASE	Ataxia-telangiectasia-like disorder 2 (ATLD2) [MIM:615919]: A neurodegenerative disorder due to defects in DNA excision repair. ATLD2 is characterized by developmental delay, ataxia, sensorineural hearing loss, short stature, cutaneous and ocular telangiectasia, and photosensitivity. {ECO:0000269|PubMed:24911150}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_FUNCTION	Auxiliary protein of DNA polymerase delta and epsilon, is involved in the control of eukaryotic DNA replication by increasing the polymerase's processibility during elongation of the leading strand (PubMed:35585232). Induces a robust stimulatory effect on the 3'-5' exonuclease and 3'-phosphodiesterase, but not apurinic-apyrimidinic (AP) endonuclease, APEX2 activities. Has to be loaded onto DNA in order to be able to stimulate APEX2. Plays a key role in DNA damage response (DDR) by being conveniently positioned at the replication fork to coordinate DNA replication with DNA repair and DNA damage tolerance pathways (PubMed:24939902). Acts as a loading platform to recruit DDR proteins that allow completion of DNA replication after DNA damage and promote postreplication repair: Monoubiquitinated PCNA leads to recruitment of translesion (TLS) polymerases, while 'Lys-63'-linked polyubiquitination of PCNA is involved in error-free pathway and employs recombination mechanisms to synthesize across the lesion (PubMed:24695737). {ECO:0000269|PubMed:18719106, ECO:0000269|PubMed:19443450, ECO:0000269|PubMed:24695737, ECO:0000269|PubMed:24939902, ECO:0000269|PubMed:35585232, ECO:0000269|PubMed:38459011}.

Main Category	Category	Terms
General Annotations	UP_COMMENT_FUNCTION	Auxiliary protein of DNA polymerase delta and epsilon, is involved in the control of eukaryotic DNA replication by increasing the polymerase's processibility during elongation of the leading strand. Induces a robust stimulatory effect on the 3'-5' exonuclease and 3'-phosphodiesterase, but not apurinic-apyrimidinic (AP) endonuclease, APEX2 activities. Has to be loaded onto DNA in order to be able to stimulate APEX2. Plays a key role in DNA damage response (DDR) by being conveniently positioned at the replication fork to coordinate DNA replication with DNA repair and DNA damage tolerance pathways. Acts as a loading platform to recruit DDR proteins that allow completion of DNA replication after DNA damage and promote postreplication repair: Monoubiquitinated PCNA leads to recruitment of translesion (TLS) polymerases, while 'Lys-63'-linked polyubiquitination of PCNA is involved in error-free pathway and employs recombination mechanisms to synthesize across the lesion. {ECO:0000256|ARBA:ARBA00045553}.
General Annotations	UP_COMMENT_FUNCTION	This protein is an auxiliary protein of DNA polymerase delta and is involved in the control of eukaryotic DNA replication by increasing the polymerase's processibility during elongation of the leading strand. {ECO:0000256|RuleBase:RU000641}.
General Annotations	UP_COMMENT_INTERACTION	P12004; P04075: ALDOA; NbExp=3; IntAct=EBI-358311, EBI-709613; P12004; P03950: ANG; NbExp=4; IntAct=EBI-358311, EBI-525291; P12004; P07355: ANXA2; NbExp=2; IntAct=EBI-358311, EBI-352622; P12004; P61769: B2M; NbExp=3; IntAct=EBI-358311, EBI-714718; P12004; P38936: CDKN1A; NbExp=39; IntAct=EBI-358311, EBI-375077; P12004; P42771: CDKN2A; NbExp=8; IntAct=EBI-358311, EBI-375053; P12004; Q9H211: CDT1; NbExp=2; IntAct=EBI-358311, EBI-456953; P12004; Q13111: CHAF1A; NbExp=4; IntAct=EBI-358311, EBI-1020839; P12004; Q96DZ9-2: CMTM5; NbExp=3; IntAct=EBI-358311, EBI-11522780; P12004; P68104: EEF1A1; NbExp=2; IntAct=EBI-358311, EBI-352162; P12004; P06733: ENO1; NbExp=3; IntAct=EBI-358311, EBI-353877; P12004; P39748: FEN1; NbExp=22; IntAct=EBI-358311, EBI-707816; P12004; P24522: GADD45A; NbExp=3; IntAct=EBI-358311, EBI-448167; P12004; P04406: GAPDH; NbExp=3; IntAct=EBI-358311, EBI-354056; P12004; Q14527: HLTF; NbExp=2; IntAct=EBI-358311, EBI-1045161; P12004; Q9H160: ING2; NbExp=3; IntAct=EBI-358311, EBI-389787; P12004; Q9H063: MAF1; NbExp=4; IntAct=EBI-358311, EBI-720354; P12004; P20585: MSH3; NbExp=6; IntAct=EBI-358311, EBI-1164205; P12004; O95944: NCR2; NbExp=7; IntAct=EBI-358311, EBI-14058375; P12004; P61970: NUTF2; NbExp=3; IntAct=EBI-358311, EBI-591778; P12004; Q15004: PCLAF; NbExp=7; IntAct=EBI-358311, EBI-10971436; P12004; P12004: PCNA; NbExp=11; IntAct=EBI-358311, EBI-358311; P12004; P18669: PGAM1; NbExp=2; IntAct=EBI-358311, EBI-717905; P12004; P00558: PGK1; NbExp=2; IntAct=EBI-358311, EBI-709599; P12004; P28340: POLD1; NbExp=3; IntAct=EBI-358311, EBI-716569; P12004; P49005: POLD2; NbExp=3;

Main Category	Category	Terms
		IntAct=EBI-358311, EBI-372354; P12004; Q15054: POLD3; NbExp=8; IntAct=EBI-358311, EBI-864956; P12004; Q9HCU8: POLD4; NbExp=4; IntAct=EBI-358311, EBI-864968; P12004; P30041: PRDX6; NbExp=2; IntAct=EBI-358311, EBI-2255129; P12004; Q9UBF6: RNF7; NbExp=3; IntAct=EBI-358311, EBI-398632; P12004; Q86TU7: SETD3; NbExp=3; IntAct=EBI-358311, EBI-2908049; P12004; Q8IWU4: SLC30A8; NbExp=3; IntAct=EBI-358311, EBI-10262251; P12004; Q9NZD8: SPG21; NbExp=3; IntAct=EBI-358311, EBI-742688; P12004; A2RU14: TMEM218; NbExp=4; IntAct=EBI-358311, EBI-10173151; P12004; P60174: TPI1; NbExp=2; IntAct=EBI-358311, EBI-717475; P12004; Q5FWF4: ZRANB3; NbExp=8; IntAct=EBI-358311, EBI-13954615; P12004; Q8TE30; NbExp=6; IntAct=EBI-358311, EBI-8874509; P12004; Q9Z111: Gadd45g; Xeno; NbExp=9; IntAct=EBI-358311, EBI-1173616; P12004; Q1K9H5: PB1; Xeno; NbExp=2; IntAct=EBI-358311, EBI-6050669; P12004; B4URF7: PB2; Xeno; NbExp=2; IntAct=EBI-358311, EBI-6050648;
General Annotations	UP_COMMENT_INTERACTION	Q6FI35; P38936: CDKN1A; NbExp=2; IntAct=EBI-8469539, EBI-375077; Q6FI35; Q9Y253: POLH; NbExp=3; IntAct=EBI-8469539, EBI-2827270; Q6FI35; Q9UNA4: POLI; NbExp=2; IntAct=EBI-8469539, EBI-741774;
General Annotations	UP_COMMENT_MISCELLANEOUS	Antibodies against PCNA are present in sera from patients with systemic lupus erythematosus.
General Annotations	UP_COMMENT_PTM	Acetylated by CREBBP and p300/EP300; preferentially acetylated by CREBBP on Lys-80, Lys-13 and Lys-14 and on Lys-77 by p300/EP300 upon loading on chromatin in response to UV irradiation (PubMed:19419956, PubMed:24939902). Lysine acetylation disrupts association with chromatin, hence promoting PCNA ubiquitination and proteasomal degradation in response to UV damage in a CREBBP- and EP300-dependent manner (PubMed:24939902). Acetylation disrupts interaction with NUDT15 and promotes degradation (PubMed:19419956). {ECO:0000269|PubMed:24939902}.
General Annotations	UP_COMMENT_PTM	Methylated on glutamate residues by ARMT1/C6orf211. {ECO:0000269|PubMed:25732820}.
General Annotations	UP_COMMENT_PTM	Phosphorylated. Phosphorylation at Tyr-211 by EGFR stabilizes chromatin-associated PCNA. {ECO:0000269|PubMed:17115032}.
General Annotations	UP_COMMENT_PTM	Ubiquitinated (PubMed:20227374, PubMed:24939902). Following DNA damage, can be either monoubiquitinated to stimulate direct bypass of DNA lesions by specialized DNA polymerases or polyubiquitinated to promote recombination-dependent DNA synthesis across DNA lesions by template switching mechanisms. Following induction of replication stress, monoubiquitinated by the UBE2B-RAD18 complex on Lys-164, leading to recruit translesion (TLS) polymerases, which are able to synthesize across DNA lesions in a potentially error-

Main Category	Category	Terms
		prone manner. An error-free pathway also exists and requires non-canonical polyubiquitination on Lys-164 through 'Lys-63' linkage of ubiquitin moieties by the E2 complex UBE2N-UBE2V2 and the E3 ligases, HLTF, RNF8 and SHPRH. This error-free pathway, also known as template switching, employs recombination mechanisms to synthesize across the lesion, using as a template the undamaged, newly synthesized strand of the sister chromatid. Monoubiquitination at Lys-164 also takes place in undamaged proliferating cells, and is mediated by the DCX(DTL) complex, leading to enhance PCNA-dependent translesion DNA synthesis. Sumoylated during S phase. {ECO:0000269|PubMed:15149598, ECO:0000269|PubMed:17108083, ECO:0000269|PubMed:17130289, ECO:0000269|PubMed:18316726, ECO:0000269|PubMed:18719106, ECO:0000269|PubMed:18948756, ECO:0000269|PubMed:20129063, ECO:0000269|PubMed:20227374, ECO:0000269|PubMed:22153967, ECO:0000269|PubMed:24939902}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the PCNA family. {ECO:0000256|ARBA:ARBA00010462, ECO:0000256|RuleBase:RU003671}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the PCNA family. {ECO:0000305}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000256|ARBA:ARBA00004123, ECO:0000256|RuleBase:RU000641}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000269|PubMed:15543136, ECO:0000269|PubMed:24115439, ECO:0000269|PubMed:24939902, ECO:0000269|PubMed:38459011}. Note=Colocalizes with CREBBP, EP300 and POLD1 to sites of DNA damage (PubMed:24939902). Forms nuclear foci representing sites of ongoing DNA replication and vary in morphology and number during S phase (PubMed:15543136). Co-localizes with SMARCA5/SNF2H and BAZ1B/WSTF at replication foci during S phase (PubMed:15543136). Together with APEX2, is redistributed in discrete nuclear foci in presence of oxidative DNA damaging agents. {ECO:0000269|PubMed:15543136, ECO:0000269|PubMed:24939902}.
General Annotations	UP_COMMENT_SUBUNIT	(Microbial infection) Interacts with herpes virus 8 protein LANA1. {ECO:0000269|PubMed:26223641}.
General Annotations	UP_COMMENT_SUBUNIT	Homotrimer (PubMed:24939902). Interacts with p300/EP300; the interaction occurs on chromatin in UV-irradiated damaged cells (PubMed:24939902). Interacts with CREBBP (via transactivation domain and C-terminus); the interaction occurs on chromatin in UV-irradiated damaged cells (PubMed:24939902). Directly interacts with POLD1, POLD3 and POLD4 subunits of the DNA polymerase delta complex, POLD3 being the major interacting partner; the interaction with POLD3 is inhibited by CDKN1A/p21(CIP1) (PubMed:11595739, PubMed:16510448, PubMed:22148433, PubMed:24939902). Forms a complex with activator 1

Main Category	Category	Terms
		heteropentamer in the presence of ATP. Interacts with EXO1, POLH, POLK, DNMT1, ERCC5, FEN1, CDC6 and POLDIP2 (PubMed:11784855, PubMed:12522211, PubMed:15149598, PubMed:15225546, PubMed:15616578, PubMed:24911150, PubMed:26760506, PubMed:9302295, PubMed:9305916, PubMed:9566895). Interacts with POLB (PubMed:19336415, PubMed:26760506). Interacts with APEX2; this interaction is triggered by reactive oxygen species and increased by misincorporation of uracil in nuclear DNA (PubMed:11376153, PubMed:19443450). Forms a ternary complex with DNTTIP2 and core histone (PubMed:12786946). Interacts with KCTD10 and PPP1R15A (By similarity). Interacts with SMARCA5/SNF2H (PubMed:15543136). Interacts with BAZ1B/WSTF; the interaction is direct and is required for BAZ1B/WSTF binding to replication foci during S phase (PubMed:15543136). Interacts with HLTF and SHPRH (PubMed:17130289, PubMed:18316726, PubMed:18719106). Interacts with NUDT15; this interaction is disrupted in response to UV irradiation and acetylation (PubMed:19419956). Interacts with CDKN1A/p21(CIP1) and CDT1; interacts via their PIP-box which also recruits the DCX(DTL) complex. The interaction with CDKN1A inhibits POLD3 binding (PubMed:11595739, PubMed:16949367, PubMed:18703516, PubMed:18794347). Interacts with DDX11 (PubMed:18499658). Interacts with EGFR; positively regulates PCNA (PubMed:17115032). Interacts with PARPBP (PubMed:22153967). Interacts (when ubiquitinated) with SPRTN; leading to enhance RAD18-mediated PCNA ubiquitination (PubMed:22681887, PubMed:27084448). Interacts (when polyubiquitinated) with ZRANB3 (PubMed:22704558, PubMed:22705370, PubMed:22759634). Interacts with SMARCAD1 (PubMed:21549307). Interacts with CDKN1C (PubMed:22634751). Interacts with PCLAF (via PIP-box) (PubMed:21628590, PubMed:23000965). Interacts with RTEL1 (via PIP-box); the interaction is direct and essential for the suppression of telomere fragility (PubMed:24115439). Interacts with FAM111A (via PIP-box); the interaction is direct and required for PCNA loading on chromatin binding (PubMed:24561620). Interacts with LIG1 (PubMed:24911150). Interacts with SETMAR (PubMed:20457750). Interacts with ANKRD17 (PubMed:23711367). Interacts with FBXO18/FBH1 (via PIP-box); the interaction recruits the DCX(DTL) complex and promotes ubiquitination and degradation of FBXO18/FBH1 (PubMed:23677613). Interacts with POLN (PubMed:19995904). Interacts with SDE2 (via PIP-box); the interaction is direct and prevents ultraviolet light induced monoubiquitination (PubMed:27906959). Component of the replisome complex composed of at least DONSON, MCM2, MCM7, PCNA and TICRR; interaction at least with PCNA occurs during DNA replication (PubMed:28191891). Interacts with MAPK15; the interaction is chromatin binding dependent and prevents MDM2-mediated PCNA destruction by inhibiting the association of PCNA with

Main Category	Category	Terms
		MDM2 (PubMed:20733054). Interacts with PARP10 (via PIP-box) (PubMed:24695737). Interacts with DDI2 (PubMed:29290612). Interacts with HMCES (via PIP-box) (PubMed:30554877). Interacts with TRAIP (via PIP-box) (PubMed:26711499, PubMed:27462463). Interacts with UHRF2 (PubMed:28951215). Interacts with ALKBH2; this interaction is enhanced during the S-phase of the cell cycle. Interacts with ATAD5; the interaction promotes USP1-mediated PCNA deubiquitination (PubMed:20147293). Interacts with DNA damage up-regulated protein DDUP (PubMed:35849344). Interacts (when phosphorylated) with GRB2 (PubMed:38459011). Interacts with ANG (PubMed:37218877). Interacts with nuclear UNG (isoform 2); this interaction mediates UNG recruitment to S-phase replication foci. Interacts with ERCC6L2 (via an atypical PIP- box); this interaction facilitates cenrtomeric localization of ERCC6L2 (PubMed:37014751). {ECO:0000250|UniProtKB:P04961, ECO:0000250|UniProtKB:P17918, ECO:0000269|PubMed:11376153, ECO:0000269|PubMed:11595739, ECO:0000269|PubMed:11784855, ECO:0000269|PubMed:12522211, ECO:0000269|PubMed:12786946, ECO:0000269|PubMed:15149598, ECO:0000269|PubMed:15225546, ECO:0000269|PubMed:15543136, ECO:0000269|PubMed:15616578, ECO:0000269|PubMed:16510448, ECO:0000269|PubMed:16949367, ECO:0000269|PubMed:17115032, ECO:0000269|PubMed:17130289, ECO:0000269|PubMed:18316726, ECO:0000269|PubMed:18499658, ECO:0000269|PubMed:18703516, ECO:0000269|PubMed:18719106, ECO:0000269|PubMed:18794347, ECO:0000269|PubMed:19336415, ECO:0000269|PubMed:19419956, ECO:0000269|PubMed:19443450, ECO:0000269|PubMed:19995904, ECO:0000269|PubMed:20147293, ECO:0000269|PubMed:20457750, ECO:0000269|PubMed:20733054, ECO:0000269|PubMed:21549307, ECO:0000269|PubMed:21628590, ECO:0000269|PubMed:22148433, ECO:0000269|PubMed:22153967, ECO:0000269|PubMed:22521144, ECO:0000269|PubMed:22634751, ECO:0000269|PubMed:22681887, ECO:0000269|PubMed:22704558, ECO:0000269|PubMed:22705370, ECO:0000269|PubMed:22759634, ECO:0000269|PubMed:23000965, ECO:0000269|PubMed:23677613, ECO:0000269|PubMed:23711367, ECO:0000269|PubMed:24115439, ECO:0000269|PubMed:24561620, ECO:0000269|PubMed:24695737, ECO:0000269|PubMed:24911150, ECO:0000269|PubMed:24939902, ECO:0000269|PubMed:26408825, ECO:0000269|PubMed:26760506, ECO:0000269|PubMed:27084448, ECO:0000269|PubMed:27462463, ECO:0000269|PubMed:27906959, ECO:0000269|PubMed:28191891, ECO:0000269|PubMed:29290612, ECO:0000269|PubMed:30554877, ECO:0000269|PubMed:35849344, ECO:0000269|PubMed:37014751, ECO:0000269|PubMed:37218877, ECO:0000269|PubMed:38459011, ECO:0000269|PubMed:9302295, ECO:0000269|PubMed:9305916, ECO:0000269|PubMed:9566895}.
Identifiers	ENTREZ_GENE_ID	5111

Main Category	Category	Terms
Identifiers	ENSEMBL_GENE_ID	ENSG00000132646
Identifiers	UNIPROT_ID	Q7Z6A0_HUMAN
Identifiers	UNIPROT_ID	Q7Z6A2_HUMAN
Identifiers	UNIPROT_ID	Q7Z6A3_HUMAN
Identifiers	UNIPROT_ID	Q7Z6A1_HUMAN
Identifiers	UNIPROT_ID	Q6FHF5_HUMAN
Identifiers	UNIPROT_ID	Q6FI35_HUMAN
Identifiers	UNIPROT_ID	PCNA_HUMAN
Interactions	DRUGBANK	Liothyronine
Interactions	DRUGBANK	Acetylsalicylic acid
Interactions	PUBCHEM	Liothyronine
Interactions	PUBCHEM	(S)-1-(1-Hydroxy-3-(4-(4-hydroxyphenoxy)-3,5-diiodophenyl)propan-2-yl)-3-isopropylurea
Interactions	PUBCHEM	3-[4-(4-Hydroxyphenoxy)-3,5-diiodophenyl]-1-morpholin-4-ylpropan-1-one
Interactions	PUBCHEM	N-[(2S)-2-amino-3-[4-(4-hydroxyphenoxy)-3,5-diiodophenyl]propyl]-3-morpholin-4-ylpropanamide
Interactions	PUBCHEM	N-[(2S)-2-amino-3-[4-(4-hydroxyphenoxy)-3,5-diiodophenyl]propyl]methanesulfonamide
Interactions	PUBCHEM	N-[(2S)-2-amino-3-[4-(4-hydroxyphenoxy)-3,5-diiodophenyl]propyl]acetamide
Interactions	PUBCHEM	N-(2-hydroxyethyl)-3-[4-(4-hydroxyphenoxy)-3,5-diiodophenyl]propanamide
Interactions	PUBCHEM	N-[(2S)-2-amino-3-[4-(4-hydroxyphenoxy)-3,5-diiodophenyl]propyl]morpholine-4-carboxamide
Interactions	PUBCHEM	4-[4-[(2S)-2-amino-3-hydroxypropyl]-2,6-diiodophenoxy]phenol
Interactions	PUBCHEM	2-[4-(4-hydroxy-3-iodophenoxy)-3,5-diiodophenyl]-N-methylacetamide
Interactions	PUBCHEM	2-[4-(4-Hydroxy-3-iodophenoxy)-3,5-diiodophenyl]-1-morpholin-4-ylethanone
Interactions	PUBCHEM	3-[4-(4-hydroxyphenoxy)-3,5-diiodophenyl]-N-(oxolan-2-ylmethyl)propanamide
Interactions	PUBCHEM	2-[4-(4-hydroxy-3-iodophenoxy)-3,5-diiodophenyl]-N,N-dimethylacetamide
Interactions	PUBCHEM	(S)-N-(1-Hydroxy-3-(4-(4-hydroxyphenoxy)-3,5-diiodophenyl)propan-2-yl)acetamide
Interactions	PUBCHEM	N-(benzenesulfonyl)-2-[4-(4-hydroxy-3-iodophenoxy)-3,5-diiodophenyl]acetamide

Main Category	Category	Terms
Interactions	PUBCHEM	N-(2-ethoxyethyl)-3-[4-(4-hydroxyphenoxy)-3,5-diiodophenyl]propanamide
Interactions	PUBCHEM	(S)-Methyl (1-Hydroxy-3-(4-(4-hydroxyphenoxy)-3,5-diiodophenyl)propan-2-yl)carbamate
Interactions	PUBCHEM	N-[(2S)-1-hydroxy-3-[4-(4-hydroxyphenoxy)-3,5-diiodophenyl]propan-2-yl]methanesulfonamide
Interactions	PUBCHEM	1-[(2S)-2-amino-3-[4-(4-hydroxyphenoxy)-3,5-diiodophenyl]propyl]-3-propan-2-ylurea
Interactions	PUBCHEM	3,5-Diiodothyronine, L-
Literature	GENERIF_SUMMARY	PCNA physically interacts with and directly binds AP endonuclease 1 (APE1) in whole cell extracts, thus demonstrating involvement of multiprotein reactions during long-patch base excision repair.
Literature	GENERIF_SUMMARY	It is hypothesized from these observations that pol epsilon and PCNA have separate but associated functions early in S phase and that pol epsilon participates with PCNA in DNA replication late in S phase.
Literature	GENERIF_SUMMARY	Functional interaction of MutY homolog with proliferating cell nuclear antigen in fission yeast (and in human)
Literature	GENERIF_SUMMARY	The over-expression of PCNA in the lens epithelium of fetus and children suggests that PCNA might be related to the development of cataract.
Literature	GENERIF_SUMMARY	The promoter of the human proliferating cell nuclear antigen gene is not sufficient for cell cycle-dependent regulation in organotypic cultures of keratinocytes
Literature	GENERIF_SUMMARY	association with histone deacetylase activity, integrating DNA replication and chromatin modification
Literature	GENERIF_SUMMARY	Human cell DNA replication is mediated by a discrete multiprotein complex. The eluted complex contains DNA polymerase delta, proliferating cell nuclear antigen, and replication protein A.
Literature	GENERIF_SUMMARY	the interaction of PCNA with DNA polymerase delta is mediated through the small subunit of the enzyme
Literature	GENERIF_SUMMARY	Expression of c-erbB-2 in node negative breast cancer does not correlate with estrogen receptor status, predictors of hormone responsiveness, or PCNA expression.
Literature	GENERIF_SUMMARY	presence in CNS tumors correlates with histological type and grade
Literature	GENERIF_SUMMARY	RAD6-dependent DNA repair is linked to modification of PCNA by ubiquitin and SUMO.
Literature	GENERIF_SUMMARY	expressive quantity of PCNA was more in hypertrophic scars and less in chronic skin ulcers

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	PCNA protein expression may help evaluate the malignancy of hypopharyngeal carcinoma and make prognostic judgement of the patients.
Literature	GENERIF_SUMMARY	relationship between the mast cell density and the context of clinicopathological parameters and expression of p185, estrogen receptor, and proliferating cell nuclear antigen in gastric carcinoma
Literature	GENERIF_SUMMARY	clinical significance of E-cadherin and proliferating cell nuclear antigen expression in adenoid cystic carcinoma
Literature	GENERIF_SUMMARY	abnormal expressions of COX-2, p53, PCNA, and nm23 associate with malignant potential, lymph node metastasis and clinical stage, and they might therefore play a role in development of gastric cancer
Literature	GENERIF_SUMMARY	The content variation of Zn, Se, Mo, Cu, Ni could be possessed of certain effect on p53 mutation and PCNA overexpression of esophageal epithelium in the high risk area.
Literature	GENERIF_SUMMARY	beta-catenin, p53 and PCNA may play important roles in the carcinogenesis of colorectal adenoma.
Literature	GENERIF_SUMMARY	Expression of p57kip2, Rb protein and PCNA and their relationships with clinicopathology in human pancreatic cancer.
Literature	GENERIF_SUMMARY	HBV infection may increase expression of PCNA and GST-pi.
Literature	GENERIF_SUMMARY	PCNA binds to two distinct functional sites on WRN
Literature	GENERIF_SUMMARY	the E2F site in the human PCNA 5' promoter is a serum-responsive element
Literature	GENERIF_SUMMARY	PCNA expression was significantly higher in recurrent aphthous ulcers than in normal oral mucosal tissue.
Literature	GENERIF_SUMMARY	Upergulation of PCNA by overexpression of tuberous sclerosis gene products.
Literature	GENERIF_SUMMARY	PARP-1 and p21 could cooperate in regulating the functions of PCNA during DNA replication/repair.
Literature	GENERIF_SUMMARY	results demonstrate for the first time a novel role for human proliferating cell nuclear antigen(PCNA)in transcriptional repression and in modulating chromatin modification with reciprocal modulation of p300 acetyltransferase and PCNA by each other
Literature	GENERIF_SUMMARY	combination of protein p53 induction and ionizing radiation activate the proliferating cell nuclear antigen gene
Literature	GENERIF_SUMMARY	The expression of PCNA is associated with the increased malignancy.
Literature	GENERIF_SUMMARY	Expression of PCNA was significantly higher than in normal pancreatic tissue. Correlated with the histological grade. Expression rate consistent with

Main Category	Category	Terms
		exacerbation of cancer. Also correlated significantly with prognosis and p53 expression.
Literature	GENERIF_SUMMARY	in hepatocellular carcinoma, PCNA participates both in DNA synthesis and repair and that highly proliferating cancers may display a sustained DNA-repair.
Literature	GENERIF_SUMMARY	Detection of PCNA and CD44mRNA expression in colorectal cancer may be useful for evaluating liver metastasis of cancer cells.
Literature	GENERIF_SUMMARY	These results suggest that PCNA trimers bound to DNA during the S phase are organized as distinct pools able to bind selectively different partners. Among them, p125-pol delta and DNA ligase I interact with PCNA in a mutually exclusive manner
Literature	GENERIF_SUMMARY	The overall expression of PCNA in all stages of fetal testis development was higher than pRb1 expression.
Literature	GENERIF_SUMMARY	proliferating cell nuclear antigen is required in different ways in 5' and 3' nick-directed excision in human mismatch repair
Literature	GENERIF_SUMMARY	PCNA expression in bone marrow plasma cells constitutes a good prognostic indicator of disease activity in patients with multiple myeloma.
Literature	GENERIF_SUMMARY	Expression of this apoptosis-related protein may be a useful marker in cervix cancer development.
Literature	GENERIF_SUMMARY	XRCC1 co-localizes with proliferating cell nuclear antigen (PCNA) at DNA replication foci, observed exclusively in the S phase of undamaged HeLa cells
Literature	GENERIF_SUMMARY	Monoubiquitinated but not unmodified PCNA interacts with polymerase eta, and we found two motifs in poleta involved in this interaction. Our findings provide a mechanism by which monoubiquitination of PCNA might mediate the polymerase switch
Literature	GENERIF_SUMMARY	In aids encephalopathy, although virtually all cells positive for HIV-1 p24 were PCNA+, there were many PCNA+ cells where HIV-1 p24 expression was not detected.
Literature	GENERIF_SUMMARY	A marker of human esophageal neoplastic progression
Literature	GENERIF_SUMMARY	Data show that an interaction between human DNA polymerase iota (poliota) and the proliferating cell nuclear antigen (PCNA) stimulates the processivity of poliota in a template-dependent manner in vitro.
Literature	GENERIF_SUMMARY	search for the factors interacting with NTH1 shows GST-NTH1 fusion protein precipitates proliferating cell nuclear antigen (PCNA) and p53 as well as XPG from human cell-free extracts
Literature	GENERIF_SUMMARY	Downregulated by cetrorelix in cultured leiomyoma cells.
Literature	GENERIF_SUMMARY	Williams syndrome transcription factor (WSTF) is

Main Category	Category	Terms
		targeted to replication foci through direct interaction with the DNA clamp PCNA
Literature	GENERIF_SUMMARY	PCNA expression induces apoptosis by up-regulating cleaved caspase-3 and Poly ADP ribose polymerasse expression and down-regulating Bcl-2 protein expression in leiomyoma cells.
Literature	GENERIF_SUMMARY	Human PCNA, either mutated to contain lysine (K) at position 85 or bearing similar primary mutations, would promote more secondary mutagenesis in cells and/or tissues where PCNA is normally expressed at low levels relative to pol delta.
Literature	GENERIF_SUMMARY	The structural basis of the FEN1-PCNA interaction was revealed by the crystal structure of the complex between human flap structure-specific endonuclease 1 and PCNA
Literature	GENERIF_SUMMARY	Expression of HPV16 E6 ORF and PCNA is significantly related to cervical carcinoma.
Literature	GENERIF_SUMMARY	Expression of Ki-67, PCNA proteins were closely connected with the high grade of stomach tumour malignancy.
Literature	GENERIF_SUMMARY	Evaluation of expression of PCNA and Ki-67 may provide an additional information about the progression of uveal melanoma.
Literature	GENERIF_SUMMARY	Rb, p21Cip1, and PCNA have roles in cAMP-mediated inhibition of DNA replication and S phase progression
Literature	GENERIF_SUMMARY	Data show that a 20-mer sequence from p21WAF1 can be associated simultaneously with PCNA and cyclin-dependent kinase/cyclin complexes.
Literature	GENERIF_SUMMARY	There is a marked loss or absence of p57(kip2) expression and high expression of PCNA in hepatocellular carcinoma
Literature	GENERIF_SUMMARY	higher expression in atrophic oral lichen planus and patients with areca quid chewing
Literature	GENERIF_SUMMARY	Limited activation of the PCNA promoter by p53 and its modified forms would restrict the amount of PCNA made available for DNA repair.
Literature	GENERIF_SUMMARY	Data supports a model that chromatin remodeling is coupled with DNA replication (and repair) through a PCNA double trimer complex.
Literature	GENERIF_SUMMARY	PCNA expression was significantly higher in bFGF positive group than in bFGF negative group in ovarian neoplasm.
Literature	GENERIF_SUMMARY	Glut-1, HK-II, and PCNA expression are related to uptake of 18F-FDG uptake in lung cancer
Literature	GENERIF_SUMMARY	show that there is a direct interaction between p66 and PCNA in living cells during DNA replication. The dominant negative effect upon growth resulting from

Main Category	Category	Terms
		expression of p66 sub-domains confirms that the p66-PCNA interaction is essential in vivo.
Literature	GENERIF_SUMMARY	YB-1 expression is correlated with PCNA expression in NSCLC. The DNA repair pathway and tumor proliferation mediated by YB-1 linking to PCNA may be responsible for controlling the growth of NSCLC.
Literature	GENERIF_SUMMARY	The decreased expression of p57(kip2) and/or overexpression of cyclinE protein and PCNA may contribute to the occurrence and progression of pancreatic cancer.
Literature	GENERIF_SUMMARY	A slight, statistically insignificant tendency towards negative correlation between PPARgamma and PCNA was found in somatotropinomas, prolactinomas, corticotropinomas and gonadotropinomas
Literature	GENERIF_SUMMARY	The induction of apoptosis coupled with the progression of DNA replication proceeds through the action of PCNA.
Literature	GENERIF_SUMMARY	PCNA and the Rad9/Rad1/Hus1 complex can independently bind and activate Fen1; acetylation of Fen1 by p300-HAT abolished the stimulatory effect of the complex but not that of PCNA, suggesting a possible mechanism of regulation of this repair pathway
Literature	GENERIF_SUMMARY	Here we use purified preparations of UNG2 and PCNA to demonstrate that UNG2 physically associates with PCNA.
Literature	GENERIF_SUMMARY	The identification of soluble PCNA as an essential replication initiation factor is reported.
Literature	GENERIF_SUMMARY	Changes in the position of PCNA result from de novo assembly of freely mobile replication factors in the nucleoplasmic pool and indicate different binding affinities for PCNA in DNA replication and repair.
Literature	GENERIF_SUMMARY	Expression of PCNA was highly correlated with the stage of nasopharyngeal carcinoma.
Literature	GENERIF_SUMMARY	Data show that there was no difference of PCNA expression in vascular smooth muscle cells of aorta and pulmonary artery trunk between hypoxia pulmonary hypertension groups and control group.
Literature	GENERIF_SUMMARY	PCNA is involved in mediating Cdt1 degradation by the Cul4-Ddb1 ligase in response to DNA damage.
Literature	GENERIF_SUMMARY	PCNA serves to promote the targeted degradation of associated proteins Cul4/Ddb1 in S-phase or after DNA damage.
Literature	GENERIF_SUMMARY	The expression of PCNA protein was lower in cortex cataract than in nuclear cataract.
Literature	GENERIF_SUMMARY	we identified ZONAB-responsive elements in PCNA promoter, PCNA expression is regulated by ZONAB

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	PCNA positive expression was closely linked with the risk of laryngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	These results indicate that early recruitment of p21 protein to DNA-damage sites is a NER-related process dependent on interaction with PCNA, thus suggesting a direct involvement of p21 in DNA repair.
Literature	GENERIF_SUMMARY	Expression of HIF-1alpha mRNA was closely correlated with the expression of survivin, PCNA, and CD34 proteins in pancreatic cancer.
Literature	GENERIF_SUMMARY	adenomas which express more than one hormone (plurihormonal adenomas) seem to have a higher PCNA indices than monohormonal ones
Literature	GENERIF_SUMMARY	PCNA, L2DTL and the DDB1-CUL4A complex play critical and differential roles in regulating the protein stability of p53 and MDM2/HDM2 in unstressed and stressed cells.
Literature	GENERIF_SUMMARY	L2DTL and PCNA interact with CUL4/DDB1 complexes and are involved in CDT1 degradation after DNA damage.
Literature	GENERIF_SUMMARY	Telomere length in alveolar type II cells and endothelial cells was significantly shorter in the patients with emphysema than in the asymptomatic nonsmokers. The level of p16INK4a expression was negatively correlated with the level of PCNA expression.
Literature	GENERIF_SUMMARY	We show the presence of residual ubiquitination of PCNA at K164 in the absence of functional RAD18, suggesting the presence of an alternate PCNA ubiquitinating enzyme in DT40.
Literature	GENERIF_SUMMARY	methyl esterification of cancer-specific PCNA represents a novel type of post-translational modification in mammalian cells that could ultimately hold the key towards unlocking its diverse functions
Literature	GENERIF_SUMMARY	PCNA, a central matchmaker for replication-linked functions, is also crucially involved in the establishment of cohesion in S phase.
Literature	GENERIF_SUMMARY	PCNA positive cells were evidenced in basal and supra basal layers of the stratified squamous epithelium and in reduced epithelium of dental follicles
Literature	GENERIF_SUMMARY	PCNA was expressed especially in the cases with increased proliferative activity in endometrial carcinoma without a significant statistic correlation between p53 and PCNA, expression.
Literature	GENERIF_SUMMARY	SHPRH function is an important deterrent to mutagenesis and carcinogenesis in humans
Literature	GENERIF_SUMMARY	results identify a novel nuclear mechanism linking tyrosine kinase receptor function with the regulation of the PCNA sliding clamp
Literature	GENERIF_SUMMARY	The PCNA and FHIT proteins were carried out in 451

Main Category	Category	Terms
		breast lesions showing nonproliferative benign breast disease, proliferative BBD without atypia, proliferative BBD with atypia, carcinoma in situ (n=15) or invasive carcinoma.
Literature	GENERIF_SUMMARY	Characterization of the human proliferating cell nuclear antigen double trimer.
Literature	GENERIF_SUMMARY	Interfering with PCNA function by overexpression of p21 mutant, impaired in its interaction with Cdks but not with PCNA, leads to ATR-dependent Chk1 activation.
Literature	GENERIF_SUMMARY	The PCNA and VEGF expression in the endothelial cells of juvenile nasopharyngeal angiofibroma were negative, but expression was positive in hemangioma endothelial cells.
Literature	GENERIF_SUMMARY	Increase in hTERT and PCNA expression is correlated to malignant serous effusion.
Literature	GENERIF_SUMMARY	an accurate mechanism exists to reduce the deleterious consequences of oxidative DNA damage, with an important role for PCNA and RP-A in determining a functional hierarchy among different DNA pols in lesion bypass
Literature	GENERIF_SUMMARY	PCNA marker was higher in helicobacter pylori infections (HP+), and HP+ with dysplasia, and gastric adenocarcinoma.
Literature	GENERIF_SUMMARY	PCNA expession in laser-radiated HepG2 cells was studied.
Literature	GENERIF_SUMMARY	Results investigate the binding properties of hundreds of short proliferating cell nuclear antigen (PCNA)-interacting peptides to archaeal and eukaryal PCNAs.
Literature	GENERIF_SUMMARY	The expressions of hTERT, p53 and PCNA were significantly higher in cystitis glandularis patients than in controls.
Literature	GENERIF_SUMMARY	PCNA and DNA repair protein, XRCC1, have distinct kinetic properties with functional consequences for their capacity to respond to successive DNA damage events
Literature	GENERIF_SUMMARY	The strong positive rate of PCNA was significantly higher in pancreatic cancer than in normal tissues.
Literature	GENERIF_SUMMARY	POLD3, plays a crucial role in the efficient recycling of PCNA during dissociation-association cycles of pol delta during elongation phase of DNA replication.
Literature	GENERIF_SUMMARY	Combination of PTEN/p53/PCNA represent an independent prognostic factor for tumor recurrence and disease-specific survival in hepatocellular carcinoma after surgery.
Literature	GENERIF_SUMMARY	human DNA glycosylase NEIL1 interacts with proliferating cell nuclear antigen and may have a role in replication-associated repair of oxidized bases

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Literature	GENERIF_SUMMARY	Recommended use of PCNA if correlations to progression of pituitary adenoma tumor growth are wanted.
Literature	GENERIF_SUMMARY	exogenous Cdt1 induces re-replication by de-repressing endogenous Cdt1 through the titration of PCNA and cyclin; Cdt1 lacking the evolutionarily conserved region that interacts with MCM2-7 is capable of inducing re-replication
Literature	GENERIF_SUMMARY	Sharp inhibition of apoptosis and changes in the levels of cell proliferation in tumor cells were paralleled by decreased expression of peripheral benzodiazepine receptor in patients with squamous cell carcinoma & skin melanoma.
Literature	GENERIF_SUMMARY	E-cad, CD44v6 and PCNA play important roles in invasion and metastasis of non-small cell lung cancer (NSCLC).
Literature	GENERIF_SUMMARY	These results point to a high level of redundancy in the DNA damage tolerance pathway and suggest the existence of another hMMS2 variant (hMMSv) or complex that can compensate for its loss.
Literature	GENERIF_SUMMARY	The combination of Ki-67 and PCNA, specific proliferative markers of PCa, may improve the accuracy of early diagnosis of prostatic cancer.
Literature	GENERIF_SUMMARY	HLTF physically interacts with the Rad6-Rad18 and Mms2-Ubc13 ubiquitin-conjugating enzyme complexes and promotes the Lys-63-linked polyubiquitination of proliferating cell nuclear antigen at its Lys-164 residue.
Literature	GENERIF_SUMMARY	the SET8 and PCNA interaction couples H4-K20 methylation with DNA replication
Literature	GENERIF_SUMMARY	PCNA ubiquitylation at K164 of PCNA is not required in vitro for pol eta to gain access to replication complexes at forks stalled by T (wedge)T and to catalyze translesion synthesis across this dimer
Literature	GENERIF_SUMMARY	interaction of the replication clamp with other repair protein(s) accounts for the essential role of PCNA in MutSalpha-dependent mismatch repair
Literature	GENERIF_SUMMARY	PAX interacting protein 1 depletion from mammalian cells by RNA interference reduced PCNA ubiquitination in response to DNA damage
Literature	GENERIF_SUMMARY	a conserved proliferating cell nuclear antigen-interacting protein sequence in Chk1 is required for checkpoint function
Literature	GENERIF_SUMMARY	Results show that show that Chk1 regulates the DNA damage-induced ubiquitination of PCNA, which facilitates the continuous replication of damaged DNA.
Literature	GENERIF_SUMMARY	progesterone receptor expression in malignant melanoma is inversely correlated with PCNA
Literature	GENERIF_SUMMARY	substitution of Lys-110 within the NLS to an alanine

Main Category	Category	Terms
		residue did not affect PCNA nuclear localization.
Literature	GENERIF_SUMMARY	A possible correlation between overexpression of p53, proliferating cell nuclear antigen (PCNA), and c-erbB-2, and the clinicopathologic features of laryngeal squamous cell carcinoma, was investigated.
Literature	GENERIF_SUMMARY	Ki-67 and PCNA are related to proliferation activities of tumor cells in acoustic neuromas. PCNA was related with clinical growth rates.
Literature	GENERIF_SUMMARY	human hepatocellular carcinoma expresses specific PCNA isoforms compared to those found in cirrhosis, implicating a role for PCNA functional alterations in hepatocarcinogenesis
Literature	GENERIF_SUMMARY	High expression of ERCC1, FLT1, NME4 and PCNA in myelodysplastic syndrome was associated with poor prognosis and disease progression along the sequence of 5q-syndrome/RCMD/RAEB/de novo AML
Literature	GENERIF_SUMMARY	NEIL1 could also participate in strand displacement repair synthesis (long patch repair (LP-BER)) mediated by FEN-1 and stimulated by PCNA.
Literature	GENERIF_SUMMARY	HLTF and SHPRH are functional homologues of yeast Rad5 that cooperatively mediate PCNA polyubiquitination and maintain genomic stability.
Literature	GENERIF_SUMMARY	Human PCNA protein carrying the Gly178Ser substitution has been crystallized. Two crystal forms have been obtained under similar conditions.
Literature	GENERIF_SUMMARY	PCNA promotes the ubiquitylation and degradation of the CDK inhibitor p21 in cells irradiated with low dose of ultraviolet radiation
Literature	GENERIF_SUMMARY	Data suggest that DNA polymerases eta and iota transiently probe DNA/chromatin; when DNA is exposed at replication forks, the polymerase residence times increase, and this is further facilitated by the ubiquitination of PCNA.
Literature	GENERIF_SUMMARY	The abnormal expression of c-met and PCNA protein was related with the oncogenesis of gastric carcinoma.
Literature	GENERIF_SUMMARY	present a model of translesion synthesis behind the replication fork to explain the persistence of ubiquitinated PCNA
Literature	GENERIF_SUMMARY	Higher levels of ERalpha-phospho-Ser118, phospho-p44/42 MAPK, and PCNA were found in leiomyoma taken from the proliferative phase compared to patient-matched myometria.
Literature	GENERIF_SUMMARY	RNF8 readily mono-ubiquitinates PCNA in the presence of UbcH5c, and polyubiquitinates PCNA in the added presence of Ubc13/Uev1a.
Literature	GENERIF_SUMMARY	Two estrogen response element sequences near the PCNA gene are not responsible for its estrogen-enhanced expression.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The binding to PCNA via its PIP domain is a prerequisite for Poleta's ability to function in translesion synthesis in human cells.
Literature	GENERIF_SUMMARY	The specific knockdown of individual protein kinases (PKs) has enabled the identification of a number of new PKs that control the expression of PCNA and p53 in ovarian cells.
Literature	GENERIF_SUMMARY	replication protein A and proliferating cell nuclear antigen act as molecular switches to activate the DNA pol lambda- dependent highly efficient and faithful repair of A
Literature	GENERIF_SUMMARY	KCTD10 can interact with PCNA
Literature	GENERIF_SUMMARY	Analysis of PCNA expression in aggressive fibromatosis (desmoid tumor) is reported.
Literature	GENERIF_SUMMARY	PCNA negatively regulates the stability of c-Abl and thereby inhibits apoptosis in the response to DNA damage.
Literature	GENERIF_SUMMARY	Data describe the crystal structures of PCNA bound to peptides containing the noncanonical PIP-box of DNA translesion synthesis polymerases, and show that the three PIP-box peptides interact with PCNA in different ways.
Literature	GENERIF_SUMMARY	The mono-ubiquitination of Proliferating Cell Nuclear Antigen mediates the switch from replicative DNA polymerases to polymerases specialised for translesion synthesis.
Literature	GENERIF_SUMMARY	CtIP is recruited to S-phase DNA replication foci through a novel motif functioning as replication foci targeting sequence.
Literature	GENERIF_SUMMARY	Data show that the expression of IGF-IR was positively related with the expression of COX-2 and PCNA.
Literature	GENERIF_SUMMARY	Data demonstrate for the first time that PCNA is protected by this newly identified partner molecule MTH2, which is related to DNA synthesis and cell cycle progression.
Literature	GENERIF_SUMMARY	PCNA mRNA monitoring in peripheral blood may provide a warning of possible over-immunosuppression as one simple assay for immune function monitoring.
Literature	GENERIF_SUMMARY	The expressions of PCNA and Bcl-2 proteins increased significantly in the lymphocytes of patients with oral lichen planus.
Literature	GENERIF_SUMMARY	Cyclosporine A inhibits colorectal cancer proliferation probably by regulating expression levels of c-Myc, p21(WAF1/CIP1) and proliferating cell nuclear antigen.
Literature	GENERIF_SUMMARY	PCNA and VEGF expression in non-small cell lung cancer regulates angiogenesis and proliferation at the same time.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The DNA binding domain (DBD) within the hLigI catalytic fragment interacts with both PCNA and the heterotrimeric cell-cycle checkpoint clamp, hRad9-hRad1-hHus1 (9-1-1).
Literature	GENERIF_SUMMARY	Data show that DADS induces G(2)/M arrest in colon cancer SW480 cells, probably through the downregulation of PCNA, p53 and cyclin B1 and upregulation of p21(WAF1).
Literature	GENERIF_SUMMARY	Data show that disruption of LigI and PCNA interactions influences trinucleotide repeat instability.
Literature	GENERIF_SUMMARY	PCNA may function as a molecular platform to facilitate the mono-ubiquitination of FANCD2 and activation of the FA-BRCA pathway
Literature	GENERIF_SUMMARY	Data show that human CAF-1 (p150) contains two PCNA interaction peptides, and suggest that these peptides are part of the mechanism that enables CAF-1 to function behind replication forks without interfering with other PCNA-mediated processes.
Literature	GENERIF_SUMMARY	CK2 could phosphorylate TNFAIP1 in vitro and in vivo, which facilitated the distribution of TNFAIP1 in nucleus and enhanced its interaction with PCNA.
Literature	GENERIF_SUMMARY	VEGF, but not PCNA may play a role in tumor recurrence and metastasis following photodynamic therapy
Literature	GENERIF_SUMMARY	The expression of PCNA was significantly higher in the FAP carcinoma tissues than in the FAP adenoma.
Literature	GENERIF_SUMMARY	hMSH2, hMLH1 and PCNA localize to the chromatin during S-phase, and accumulate to a greater extent in cells treated with a DNA alkylating agent.
Literature	GENERIF_SUMMARY	proliferating cell nuclear antigen, ATR, TopBP1, and Chk1 are recruited to chromatin in a MutLalpha- and MutSalpha-dependent fashion after N-methyl-N'-nitro-N-nitrosoguanidine treatment.
Literature	GENERIF_SUMMARY	The authors obtained direct evidence that polymerase switching reactions are stimulated by mono-ubiquitination of PCNA, depending on a function of the ubiquitin binding zinc finger domain of pol eta.
Literature	GENERIF_SUMMARY	there is a p53-independent function of MTA1 in DNA damage response via modulation of the p21 WAF1-proliferating cell nuclear antigen pathway
Literature	GENERIF_SUMMARY	The increased gene expressions of bFGF, alpha-SMA and PCNA in atrium during atrial fibrillation may contribute to atrial fibrosis by promoting fibroblast proliferation.
Literature	GENERIF_SUMMARY	The anti-tumor mechanisms of genistein may be related with the down-regulation of expression of VEGF and PCNA, and up-regulation of the expression of p21.
Literature	GENERIF_SUMMARY	In pancreatic cancer, loss of p27 and increased p53 expression is associated with a more aggressive

Main Category	Category	Terms
		phenotype
Literature	GENERIF_SUMMARY	PCNA forms a complex with Hsp90 in cancer cell lines. Incubation with geldanamycin resulted in the degradation of both PCNA and Hsp90.
Literature	GENERIF_SUMMARY	PCNA increases the fidelity of thymine glycol bypass and does not influence normal DNA synthesis catalyzed by DNA polymerase lambda.
Literature	GENERIF_SUMMARY	Aurora A interacts through its N-terminal domain with a region of human Gadd45alpha and Gadd45alpha appears to interact only weakly with PCNA through its flexible loop.
Literature	GENERIF_SUMMARY	PCNA did not influence the elongation by DNA pol and had no effect on inhibition of elongation caused by RPA
Literature	GENERIF_SUMMARY	Treatment of HepG2 cells with HMBG1 increases expression of cyclin D1 and PCNA.
Literature	GENERIF_SUMMARY	Most nuclei of TUNEL-positive myocytes in the fibrillating right atria express proliferating cell nuclear antigen (PCNA), and most nuclei of TUNEL-positive myocytes also express SC-35; none express Ki-67.
Literature	GENERIF_SUMMARY	PCNA expression levels were significantly decreased in patient with schizophrenia.
Literature	GENERIF_SUMMARY	results reveal that the disruption of Dnmt1/PCNA/UHRF1 interactions acts as an oncogenic event and that one of its signatures (i.e. the low level of mMTase activity) is a molecular biomarker associated with a poor prognosis in GBM patients
Literature	GENERIF_SUMMARY	the methylated form, but not the phosphorylated form, of FEN1 strongly interacts with proliferating cell nuclear antigen (PCNA), ensuring the 'on' and 'off' timing of its reaction
Literature	GENERIF_SUMMARY	Data show that ERK8 prevents HDM2-mediated PCNA destruction by inhibiting the association of PCNA with HDM2, and implicate ERK8 in the regulation of genomic stability.
Literature	GENERIF_SUMMARY	PCNA immunohistochemical staining is low in fetal areas and very high in embryonal areas of hepatoblastoma. Expression level had no prognostic significance.
Literature	GENERIF_SUMMARY	PCNA and MutLalpha
Literature	GENERIF_SUMMARY	Stable interaction between the human proliferating cell nuclear antigen loader complex Ctf18-replication factor C (RFC) and DNA polymerase {epsilon} is mediated by the cohesion-specific subunits, Ctf18, Dcc1, and Ctf8.
Literature	GENERIF_SUMMARY	There was no significant difference in positive intensity of PCNA between primary and metastatic ovarian cancers.
Literature	GENERIF_SUMMARY	Data show that PCNA is associated with several

Main Category	Category	Terms
		cytoplasmic oncoproteins and enzymes in the glycolysis pathway.
Literature	GENERIF_SUMMARY	This study identifies CRL4-Cdt2 ubiquitin ligase to promote the ubiquitin-dependent proteolysis of the histone H4 methyltransferase Set8 during S-phase of the cell cycle and after UV-irradiation in a reaction that is dependent on PCNA.
Literature	GENERIF_SUMMARY	PCNA as a regulator of neutrophil lifespan.
Literature	GENERIF_SUMMARY	Increased expression of the cancer-associated isoform of PCNA is common in prostatic adenocarcinoma.
Literature	GENERIF_SUMMARY	PR-Set7 is transiently recruited to laser-induced DNA damage sites through its interaction with PCNA, after which 53BP1 is recruited dependent on PR-Set7 catalytic activity.
Literature	GENERIF_SUMMARY	The turnover of SET8 is accelerated after ultraviolet irradiation dependent on the CRL4(CDT2) ubiquitin ligase and PCNA.
Literature	GENERIF_SUMMARY	The high expression of PD4, CD44 and PCNA proteins may be closely related to the pathogenesis, development and prognosis of laryngeal carcinoma.
Literature	GENERIF_SUMMARY	Stress treatment of mouse cells with ethanol or hydrogen peroxide caused the re-distribution of MSH3 into nuclear bodies containing the proliferating cell nuclear antigen (PCNA), a known binding partner of MutSbeta.
Literature	GENERIF_SUMMARY	The increased dynamics of hPCNA relative to scPCNA may allow it to acquire multiple induced conformations upon binding to its substrates enlarging its binding diversity.
Literature	GENERIF_SUMMARY	reduced immunostaining of PCNA and Ki67 and increased immunostaining of p27 and p57 in the dexamethasone-induced intrauterine growth restriction placental development compared to control placental development
Literature	GENERIF_SUMMARY	The purpose was to correlate markers expression of apoptosis (p53 and bcl-2) and cell proliferation (Ki-67 and PCNA) with histological indicators of tumor severity.
Literature	GENERIF_SUMMARY	Studies indicate that the ubiquitination and SUMOylation of a key DNA replication/repair protein PCNA controls precisely how eukaryotic cells respond to different types of DNA damage.
Literature	GENERIF_SUMMARY	high PCNA-LI (PCNA+) was observed in 28/59 cases (47.5%). many of the p53+/PCNA+ cases could be confirmed as highly invasive cancer. p53+/PCNA+ cases showed high risk of tumor recurrence. patients with p53+/PCNA+ had a worse prognosis.
Literature	GENERIF_SUMMARY	Studies indicate that it is generally believed that monoubiquitinated PCNA promotes translesion DNA

Main Category	Category	Terms
		synthesis, whereas polyubiquitinated PCNA mediates an error-free mode of lesion bypass.
Literature	GENERIF_SUMMARY	Studies indicate that monoubiquitylation of PCNA allows mutagenic translesion synthesis by damage-tolerant DNA polymerases, polyubiquitylation is required mainly for an error-free pathway that likely involves template switching.
Literature	GENERIF_SUMMARY	Data indicate that PAF15 associates with PCNA, and depletion of PAF15 decreases the number of cells in S phase, suggesting a role for it in cell cycle regulation.
Literature	GENERIF_SUMMARY	Studies indicate that USP1 and ELG1 as regulators of PCNA ubiquitylation.
Literature	GENERIF_SUMMARY	Studies indicate that 48 biomarkers have been identified for ccRCC patients, only CD44, CA9, p53, Ki67 and PCNA have shown prognostic value, and IMP-3 and VEGF-R2 are predictors of survival of pRCC patients.
Literature	GENERIF_SUMMARY	Studies indicate that All the Y-family polymerases have ubiquitin binding domains that bind to mono-ubiquitinated PCNA to effect the switching from replicative to Y-family polymerase.
Literature	GENERIF_SUMMARY	The expression levels of PCNA, Ki-67, Bcl-2 and mP53 are significantly increased in the pterygium samples as compared to the normal conjunctiva samples.
Literature	GENERIF_SUMMARY	Studies suggest that DNA damage-induced ubiquitination or sumoylation of PCNA prevents CRL4Cdt2-dependent degradation by inhibiting binding of Cdt1 to PCNA.
Literature	GENERIF_SUMMARY	In response to oxidative stress, PCNA becomes transiently monoubiquitinated in an S phase- and USP1-independent manner.
Literature	GENERIF_SUMMARY	Parathyroid adenomas come into existence as a result of neoplastic proliferation of PCNA/Ki-67 positive chief cells, especially vacuolated chief cells.
Literature	GENERIF_SUMMARY	The 9-1-1 and proliferating cell nuclear antigen (PCNA) are compared and discussed in light of their respective structures and functions.
Literature	GENERIF_SUMMARY	We demonstrate that PCNA promotes cancer survival by immune evasion through inhibition of NKp44-mediated NK cell attack.
Literature	GENERIF_SUMMARY	The gene expressions of PCNA, c-erbB-2, and p53 gradually decreased with the increased distance from primary tumor (P 0.05).
Literature	GENERIF_SUMMARY	Genetic variants in PCNA does not contribute to the development of Lynch syndrome.
Literature	GENERIF_SUMMARY	Phosphomimetic mutation of serine-458 to aspartate decreases the binding of p68 subunit of Poldelta to PCNA

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Ki-67, but not proliferating cell nuclear antigen, has a role in poor disease free survival in patients with adenocarcinoma of the lung
Literature	GENERIF_SUMMARY	PCNA signal transduction has an important impact on the growth regulation of breast cancer cells.
Literature	GENERIF_SUMMARY	residue in PCNA that is essential to support destruction of all CRL4(Cdt2) substrates
Literature	GENERIF_SUMMARY	Statistically significant correlation was found between PCNA expression and the presence of lymphogenous metastasis of squamous cell carcinoma of the larynx.
Literature	GENERIF_SUMMARY	ATM and PCNA physically interact both in vivo and in vitro.
Literature	GENERIF_SUMMARY	[review] PCNA autoantigen is commonly associated with systemic lupus erythematosus (SLE). The primary antigenic target is a 34 kDa protein that is part of the DNA polymerase delta multi-protein complex.
Literature	GENERIF_SUMMARY	PCNA is inhibited by T2AA, a molecule that disrupts interactions with PIP-box proteins and inhibits DNA replication
Literature	GENERIF_SUMMARY	a role for H3K56me1 as a chromatin docking site for PCNA prior to its function in DNA replication.
Literature	GENERIF_SUMMARY	nucleosome binding protein HMGN1 as a new PCNA-interacting protein that enhances the binding of PCNA to chromatin
Literature	GENERIF_SUMMARY	The findings suggest the importance of SUMO modification of human PCNA in preventing replication fork collapse to DNA double-strand breaks and providing genome stability.
Literature	GENERIF_SUMMARY	lower levels of Ki-67 and PCNA expression were measured in cellular leiomyomas than in leiomyosarcomas
Literature	GENERIF_SUMMARY	Data show that the ability of checkpoint kinases Chk1 to promote replication fork progression on damaged DNA relies on its proliferating cell nuclear antigen (PCNA)-interacting motif, which is required for its release from chromatin after DNA damage.
Literature	GENERIF_SUMMARY	An increase of methylated PCNA was found in cancer cells, and the expression levels of SETD8 and PCNA were correlated in cancer tissue samples
Literature	GENERIF_SUMMARY	Data show models for the ternary PCNA/FEN1/DNA and Rad9-Rad1-Hus1 (9-1-1 complex)/FEN1/DNA assemblies.
Literature	GENERIF_SUMMARY	The ub-PCNA is impaired in its ability to support the coordinated actions of Fen1 and Pol delta in assays mimicking Okazaki fragment processing.
Literature	GENERIF_SUMMARY	PCNA and HP1 beta, a key player in DNA replication, are closely spaced components of a multiprotein complex

Main Category	Category	Terms
		involved in replication, both in S phase and during DNA repair, and that the functional complex requires formation of an HP1 dimer.
Literature	GENERIF_SUMMARY	Elevated levels of proliferating cell nuclear antigen is associated with increased aggressiveness in breast cancer.
Literature	GENERIF_SUMMARY	Epstein-Barr virus BPLF1 deubiquitinates PCNA and attenuates polymerase eta recruitment to DNA damage sites
Literature	GENERIF_SUMMARY	All IMAGe-associated mutations clustered in the PCNA-binding domain of CDKN1C and resulted in loss of PCNA binding.
Literature	GENERIF_SUMMARY	Mutation of Lys-164 to arginine prevents PCNA ubiquitylation and rescues the degradation of the K164R/Y211F PCNA double mutant
Literature	GENERIF_SUMMARY	ZRANB3 translocase, a SNF2 family member related to the SIOD disorder SMARCAL1 protein, is recruited by polyubiquitinated PCNA to promote fork restart following replication arrest.
Literature	GENERIF_SUMMARY	annealing helicase (AH)2 binds to PCNA, which is crucial for its function at stalled replication forks.
Literature	GENERIF_SUMMARY	PCNA expression was decreased but ICAM-1 expression was significantly increased in the NIDDM and cataract group compared with that of non-diabetic cataract patients
Literature	GENERIF_SUMMARY	Nuclear-to-cytoplasmic relocalization of the proliferating cell nuclear antigen (PCNA) during differentiation involves a chromosome region maintenance 1 (CRM1)-dependent export and is a prerequisite for PCNA antiapoptotic activity in mature neutrophils
Literature	GENERIF_SUMMARY	X-ray crystal structures of ubiquitin-modified and SUMO-modified PCNA have provided insights into how these post-translational modifications alter the specificity of PCNA for some of its binding partners
Literature	GENERIF_SUMMARY	Data indicate that terminal differentiation and markers of senescence PCNA and p16 in the descendants of cells traversed by one carbon ion occurred earlier than in controls.
Literature	GENERIF_SUMMARY	the ubiquitin moiety is flexible relative to the PCNA, with its hydrophobic patch mostly accessible for recruitment of TLS polymerases, which promotes the interaction with polymerase eta
Literature	GENERIF_SUMMARY	The Pol beta, but not Pol lambda, can bypass a single L lesion in the template, and its activity is increased by the auxiliary protein proliferating cell nuclear antigen (PCNA), whereas both enzymes were completely blocked by a tandem lesion.
Literature	GENERIF_SUMMARY	The interactions of PCNA with several other proteins

Main Category	Category	Terms
		and fragments in solution, were characterized.
Literature	GENERIF_SUMMARY	APIM in XPA is a functional PCNA interacting motif.
Literature	GENERIF_SUMMARY	ATAD5 regulates the cycle of DNA replication factories, probably through its PCNA-unloading activity.
Literature	GENERIF_SUMMARY	PCNA up-regulation is associated with laryngeal epitheliopapillomatous lesions.
Literature	GENERIF_SUMMARY	The HP1-PCNA complex may serve as a loading and supporting platform recruiting various factors involved in DNA replication.
Literature	GENERIF_SUMMARY	a novel non-catalytic role for POLH in promoting PCNA monoubiquitination.
Literature	GENERIF_SUMMARY	PCNA binding site on Gadd45alpha plays a critical role in modulating the interaction with PCNA and APE1, affecting base excision repair.
Literature	GENERIF_SUMMARY	Novel interaction between proliferating cell nuclear antigen and HLA I on the surface of tumor cells inhibits NK cell function through NKp44.
Literature	GENERIF_SUMMARY	findings identify the pathway of Ron-c-Abl-PCNA as a mechanism of oncogene-induced cell proliferation, with potentially important implications for development of combination therapy of breast cancer
Literature	GENERIF_SUMMARY	Posttranslational modifications of PCNA affect its activity and interaction partners. [Review]
Literature	GENERIF_SUMMARY	The presence of PCNA on the D-loop regulates the length of the extension tracks by recruiting various polymerases and might present a regulatory point for the various recombination outcomes.
Literature	GENERIF_SUMMARY	This study tested the hypothesis that monomeric PCNA could have a biological role in neutrophils.
Literature	GENERIF_SUMMARY	(CAG)n and (CTG)n extrusions in relaxed closed circular DNA do in fact support MutSbeta-, replication factor C-, and PCNA-dependent activation of MutLalpha endonuclease
Literature	GENERIF_SUMMARY	the promotion of base excision repair activity by a p53-dependent pathway is mediated by an interaction between Gadd45a, PCNA and APE1 in the presence of organic selenium that results in increased APE1 activity
Literature	GENERIF_SUMMARY	our results show that the-PCNA interaction is important in regulating DNA synthesis and contributes to tumorigenesis.
Literature	GENERIF_SUMMARY	cell cycle related proteins PCNA, Ki67, cyclin D3, p27 and p57 were expressed in both normal and diabetic human term placentas.
Literature	GENERIF_SUMMARY	PCNA rs25406 polymorphism is associated with lung cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	A significant positive correlation was found between age and PCNA and MIB-1 values (P
Literature	GENERIF_SUMMARY	Data show that YB-1 disrupts MutSalpha/PCNA/G/T heteroduplex ternary complex formation through competing with MutSa for PCNA binding on G/T heteroduplex.
Literature	GENERIF_SUMMARY	DDB2 is a PCNA-binding protein and this association is required for DDB2 proteolytic degradation.
Literature	GENERIF_SUMMARY	Data suggest that ESR1 (estrogen receptor 1) regulates cell cycle by suppressing expression of p53/p21 (tumor protein p53/cyclin-dependent kinase inhibitor 1A protein) and up-regulating proliferating cell nuclear antigen (PCNA).
Literature	GENERIF_SUMMARY	PCNA unloading is delayed in the absence of nucleosome assembly.
Literature	GENERIF_SUMMARY	Proliferating cell nuclear antigen immunoreactivity is strong and diffuse in the majority of hepatocellular carcinomas but in only a minority of benign liver nodular lesions.
Literature	GENERIF_SUMMARY	the immunohistochemical distribution of PCNA and p53 are strongly coordinated with each other in villous and extravillous cells in human placenta.
Literature	GENERIF_SUMMARY	The global DNA hypomethylation induced by the DNMT1/PCNA/UHRF1 disruption is an oncogenic event of human tumorigenesis.
Literature	GENERIF_SUMMARY	PARP10 binding to PCNA is required for translesion DNA synthesis.
Literature	GENERIF_SUMMARY	while interaction with PCNA was important for targeting p21 to the CRL4Cdt2 ligase re-localized to MVM replication centers
Literature	GENERIF_SUMMARY	Findings establish a crucial role of PCNA ISG15 modification in termination of error-prone translesion DNA synthesis for preventing excessive mutagenesis
Literature	GENERIF_SUMMARY	PCNA and Ki67 staining intensities significantly increased in villous parts, significantly decreased in basal plates of Preeclampsia group
Literature	GENERIF_SUMMARY	describe a syndrome in which the cardinal clinical features include short stature, hearing loss, premature aging, telangiectasia, neurodegeneration, and photosensitivity, resulting from a homozygous missense (p.Ser228Ile) sequence alteration of PCNA
Literature	GENERIF_SUMMARY	CRL4(Cdt2)-dependent degradation of TDG occurs in S phase because of the requirement for TDG to interact with chromatin-loaded PCNA, and this degradation is important for preventing toxicity from excess TDG.
Literature	GENERIF_SUMMARY	this is a first study showing the correlation of Ki-67, PCNA interacted with CDK1 might lead to malignant transformation

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	A reverse PIP Box interaction occurs with PCNA. Small-molecule ligand binding at the PIP Box interaction site confirmed the adaptive nature of the protein in dictating overall shape and implicates allosterism in transmitting biological effects.
Literature	GENERIF_SUMMARY	The low expression of proliferation markers such as PCNA.
Literature	GENERIF_SUMMARY	14-3-3zeta reduces DNA damage by interacting with and stabilizing proliferating cell nuclear antigen
Literature	GENERIF_SUMMARY	The aberrant DNA replication mediated by the PCNA-DNA pol-b complex induces p53-dependent neuronal cell death
Literature	GENERIF_SUMMARY	our findings reveal a modulatory role of USP7 in PCNA ubiquitination-mediated stress-tolerance pathways by fine-tuning Poleta turnover.
Literature	GENERIF_SUMMARY	The S-nitrosylation status of PCNA localized in cytosol impacts the apoptotic pathway in a Parkinson's disease paradigm
Literature	GENERIF_SUMMARY	Results revealed that mutation of lysine K164 in PCNA disrupts mono-ubiquitination of the multiple units within a homo-trimer complex as well as a DNA damage tolerance pathway.
Literature	GENERIF_SUMMARY	Mutant p53 depletion profoundly influenced PARP1 localization and increased the level of PCNA and MCM4 proteins.
Literature	GENERIF_SUMMARY	p15PAF acts as a flexible drag that regulates PCNA sliding along the DNA and facilitates the switch from replicative to translesion synthesis polymerase binding.
Literature	GENERIF_SUMMARY	PCNA phosphorylation by EGFR inhibits mismatch repair and promotes misincorporation during DNA synthesis
Literature	GENERIF_SUMMARY	the hydroxyl group of Tyr151 in p21 acts as a tethering point for ideal packing and surface recognition of the peptide interface, increasing the binding affinity of p21 for PCNA.
Literature	GENERIF_SUMMARY	The expression of BAF180 promotes UV-induced PCNA ubiquitination during S phase.
Literature	GENERIF_SUMMARY	Bub1 in complex with LANA recruits PCNA to regulate Kaposi's sarcoma-associated herpesvirus latent replication and DNA translesion synthesis.
Literature	GENERIF_SUMMARY	Report PCNA expression in giant cell tumor of bones.
Literature	GENERIF_SUMMARY	adenine glycosylase activity, mismatch recognition properties and interaction with protein partners of MUTYH and 5 MAP variants were examined; P502L and R520Q had reduced affinity for PCNA; only Q324H was found to have reduced affinity for Hus1
Literature	GENERIF_SUMMARY	Hydrophobic pocket of PCNA is the key domain

Main Category	Category	Terms
		mediating ALKBH2/PCNA interaction. PCNA association with ALKBH2 increases significantly during DNA replication, suggesting that ALKBH2 forms a cell-cycle dependent complex with PCNA.
Literature	GENERIF_SUMMARY	Studies show that acidic residue methyltransferase 1 (Armt1) has a vital role in regulation of the DNA damage response likely through its ability to O-methylate glutamyl residues of the DNA repair factor proliferating cell nuclear antigen (PCNA).
Literature	GENERIF_SUMMARY	Effect of Fenugreek seed extract (FCE) was studied using HepG2 cell line; evaluated apoptosis effect and changes in expression of PCNA and tp53 due to FCE.
Literature	GENERIF_SUMMARY	USP7 suppresses H2O2-induced mutagenesis involving cell-cycle-independent processes such as DNA repair.
Literature	GENERIF_SUMMARY	The authors describe the structure of the disease-causing S228I variant, which reveals a large conformational change that dramatically transforms the binding pocket for PCNA client proteins.
Literature	GENERIF_SUMMARY	data indicated that lncRNA-PCNA-AS1 may participate in the gastric cancer carcinogenesis and development and may serve as a new biomarker for patients with gastric cancer
Literature	GENERIF_SUMMARY	REV1 promote PCNA monoubiquitylation after UV radiation through interacting with ubiquitylated RAD18.
Literature	GENERIF_SUMMARY	Studies demonstrated that POLH interaction with PCNA plays an indispensable role in the recruitment of pol-eta to UV-induced cyclobutane pyrimidine dimers site to carry out translesion synthesis (TLS), and the dissociation of pol-eta from PCNA after TLS is essential for maintaining high fidelity DNA replication.
Literature	GENERIF_SUMMARY	These findings offer new insights into the determinants of PIP box for PCNA binding.
Literature	GENERIF_SUMMARY	The aim of this study is to analyze the immunoexpression of Ki67, p53, MCM3 and PCNA markers in epithelial remnants of dental follicles of impacted teeth and to identify a possible correlation between the immunoexpression of these markers in dentigerous cysts and keratocystic odontogenic tumors.
Literature	GENERIF_SUMMARY	Disruption of PCNA-lamins A/C interactions by prelamin A induces DNA replication fork stalling
Literature	GENERIF_SUMMARY	this study shows that cytoplasmic PCNA connects glycolysis and cell survival in acute myeloid leukemia
Literature	GENERIF_SUMMARY	PCNA plays a critical role in preventing CAG repeat expansions by modulating the structures of dynamic DNA via cooperation with base excision repair enzymes.
Literature	GENERIF_SUMMARY	collectively, this study uncovers a new role for CRL4CDT2 in protecting genomic integrity against

Main Category	Category	Terms
		replication stress via regulated proteolysis of PCNA-associated SDE2 and provides insights into how an integrated UBL domain within linear polypeptide sequence controls protein stability and function.
Literature	GENERIF_SUMMARY	Eco1 mediated acetylation regulates PCNA sliding on DNA in the presence of DNA damage, favoring homologous recombination linked to sister-chromatid cohesion.
Literature	GENERIF_SUMMARY	PCNA recognizes the DNA structure through a set of basic residues within the ring channel organized to match the pitch of B-DNA, establishing short-lived polar interactions with consecutive DNA phosphates.
Literature	GENERIF_SUMMARY	Reduced affinity of PCNA(S228I) for specific clients causes subtle cellular defects in undamaged cells which likely contribute to the etiology of PCNA-associated DNA repair disorder (PARD). Analysis of the binding between PCNA and PCNA-interacting proteins (PIPs) shows that the S228I change dramatically impairs the majority of these interactions, including that of Cdt1, DNMT1, PolD3(p66) and PolD4(p12).
Literature	GENERIF_SUMMARY	Multiple studies provide evidence that PCNA lies at the center of the faithful duplication of eukaryotic genomes and therefore genome integrity. [review]
Literature	GENERIF_SUMMARY	Replication protein A binds tightly to the single-stranded DNA adjacent to a blocked primer/template junction and restricts PCNA to the upstream duplex region by physically blocking diffusion of PCNA along ssDNA.
Literature	GENERIF_SUMMARY	The complex between p15PAF and trimeric PCNA is of low affinity, forming a transient complex that is difficult to characterize at a structural level due to its inherent polydispersity. We have determined the structure, conformational fluctuations, and relative population of the five species that coexist in solution by combining small-angle X-ray scattering (SAXS) with molecular modelling.
Literature	GENERIF_SUMMARY	Human CTF18-RFC clamp-loader complexed with non-synthesising POLE efficiently loads the PCNA sliding clamp.
Literature	GENERIF_SUMMARY	PCNA is a substrate of the HSV-1 deubiquitinase UL36USP, which has previously been shown to be involved mainly in virus uptake and maturation
Literature	GENERIF_SUMMARY	ECRG2 in combination with cisplatin (DDP) can inhibit viability and induce apoptosis in esophageal cancer DDP-resistant cells, possibly via upregulation of p53 expression and downregulation of PCNA expression.
Literature	GENERIF_SUMMARY	High PCNA expression is associated with lung adenocarcinoma.
Literature	GENERIF_SUMMARY	PCNA and MutLalpha interact specifically but weakly in solution to form a complex of approximately 1
Literature	GENERIF_SUMMARY	Treatment with 240 mg/l matrine reduced the protein

Main Category	Category	Terms
		expression levels of PCNA and eIF4E. Matrine also reduced the migration ability of A549 cells and inhibited their proliferation, which may be associated with the overexpression of p53 and p21, and the reduction of PCNA and eIF4E expression levels.
Literature	GENERIF_SUMMARY	These results propose a mechanism for progerin-induced genome instability and accelerated replicative senescence in Hutchinson-Gilford progeria syndrome.
Literature	GENERIF_SUMMARY	Data suggest that, during human DNA replication, restricting PCNA (proliferating cell nuclear antigen) to the vicinity of its DNA target site is important; PCNA can be maintained at or near primer/template junctions during DNA synthesis by RPA (replication protein A) or SSB (single-stranded DNA-binding protein); here, the SSB used was from Escherichia coli.
Literature	GENERIF_SUMMARY	PCNA and E-cadherin may have roles in prognosis of patients with gastric cancer
Literature	GENERIF_SUMMARY	Results suggest that the UNG2 N-terminus may serve as a flexible scaffold to tether PCNA and RPA at the replication fork, and that post-translational modifications on the UNG2 N-terminus disrupt formation of the PCNA-UNG2-RPA protein complex.
Literature	GENERIF_SUMMARY	The results revealed that PCNA is predominantly localized in the cytoplasm, while hAng is distributed both in the nucleus and in the cytoplasm. hAng and PCNA colocalize in the cytoplasm, suggesting that they may interact in this compartment.
Literature	GENERIF_SUMMARY	The here presented evidence that the PCNA inner surface is highly regulated to control DNA damage resistance represents a new concept that offers new opportunities to develop tools to manipulate the DNA damage response in cancer treatment. [review]
Literature	GENERIF_SUMMARY	Damage-induced fork reversal in mammalian cells requires PCNA ubiquitination, UBC13, and K63-linked polyubiquitin chains, previously involved in error-free damage tolerance. Fork reversal in vivo also requires ZRANB3 translocase activity and its interaction with polyubiquitinated PCNA, pinpointing ZRANB3 as a key effector of error-free DNA damage tolerance.
Literature	GENERIF_SUMMARY	Data suggest that, as part of DNA repair in nucleus of embryonic stem cells, IGF1R interacts with and phosphorylates PCNA at tyrosine residues 60, 133, and 250; this is followed by mono- and polyubiquitination of PCNA by RAD18 and SHPRH. (IGF1R = insulin-like growth factor 1 receptor; PCNA = proliferating cell nuclear antigen; RAD18 = E3 ubiquitin-protein ligase RAD18; SHPRH = E3 ubiquitin-protein ligase SHPRH)
Literature	GENERIF_SUMMARY	Dynamic binding of the PARG non-canonical PIP-box to PCNA.
Literature	GENERIF_SUMMARY	Here, we report the complex structure of PCNA and the peptide ((784)NEILQTLLDLFFPGYSK(800)) derived from UHRF2 that contains a PIP box. Structural analysis

Main Category	Category	Terms
		combined with mutagenesis experiments provide the molecular basis for the recognition of UHRF2 by PCNA via PIP-box.
Literature	GENERIF_SUMMARY	PCNA expression significantly associated with clinical stage, histological grade, and poor prognosis of Osteosarcoma, which could evaluate tumor cell proliferation, and predict its biological behavior and prognosis.
Literature	GENERIF_SUMMARY	FAN1 interaction with ubiquitylated PCNA alleviates replication stress and preserves genomic integrity independently of BRCA2
Literature	GENERIF_SUMMARY	This study identified PCNA-peptide interactions formed in the peptide bound simulation that play a crucial role in complex formation. The calculated binding energies correlate well with the measured binding affinities of various peptides to PCNA.
Literature	GENERIF_SUMMARY	upregulated in colorectal cancer
Literature	GENERIF_SUMMARY	CHAF1A and PCNA are highly expressed in cervical squamous cell carcinoma and associated with the malignancy
Literature	GENERIF_SUMMARY	Results suggest that CDK-mediated phosphorylation of Cdt2 inactivates its ubiquitin ligase activity by reducing its affinity to PCNA, an important strategy for regulating the levels of key proteins in the cell cycle.
Literature	GENERIF_SUMMARY	DDB2-PCNA interaction may contribute to a correct DNA damage response for maintaining genome integrity
Literature	GENERIF_SUMMARY	Results show that HECW2 interacts with PCNA mediating its ubiquitination and proteasomal degradation.
Literature	GENERIF_SUMMARY	We therefore identified the NKp44 binding site to PCNA and further developed an NKp44-peptide-based agent that can inhibit tumor growth through interfering with the function of intracellular PCNA in the tumor cell.
Literature	GENERIF_SUMMARY	It has been shown that conjugation of SUMO2, but not SUMO1 or SUMO3, to the essential replication factor PCNA is induced on transcribed chromatin by the RNAPII-bound helicase RECQ5.
Literature	GENERIF_SUMMARY	our biochemical data show that p15PAF impairs primer synthesis by pol eta-PCNA holoenzyme against both damaged and normal DNA templates. In light of our findings, we discuss the possible mechanistic roles of p15PAF in DNA replication and suppression of DNA lesion bypass.
Literature	GENERIF_SUMMARY	Alpha-fetoprotein-positive hepatocellular carcinoma patients had high CK19 levels and low PCNA levels.
Literature	GENERIF_SUMMARY	the formation of the PCNA trimeric clamp around DNA during DNA replication and repair may bring together CDT1 and CRL4(CDT2) ubiquitin E3 ligase to target

Main Category	Category	Terms
		CDT1 for proteolysis in a DNA synthesis-dependent manner.
Literature	GENERIF_SUMMARY	Regulation of HLTF-mediated PCNA polyubiquitination by RFC and PCNA monoubiquitination levels determines choice of damage tolerance pathway.
Literature	GENERIF_SUMMARY	the complexity of replication fork composition upon DNA damage was examined by using a PCNA-based proteomic screen to uncover known and unexplored players involved in replication and replication stress response.
Literature	GENERIF_SUMMARY	The USP7 efficiently removes polyubiquitin chains from polyubiquitinated PCNA by preferential cleavage of the PCNA-ubiquitin linkage.
Literature	GENERIF_SUMMARY	RPA inhibits monoubiquitination of free PCNA by directly interacting with Rad18.
Literature	GENERIF_SUMMARY	Identifies AKT as a novel regulator of PCNA ubiquitylation.
Literature	GENERIF_SUMMARY	We found that VASP's interaction with Pfn1 is promoted by cell-substrate adhesion and requires down-regulation of PKA activity
Literature	GENERIF_SUMMARY	CCEPR is a novel clinical biomarker for prognosis and regulates cell proliferation through PCNA in osteosarcoma.
Literature	GENERIF_SUMMARY	Based on the increased expression of PCNA and the development of the hypoxia process in the presence of extracellular matrix, our study showed that the addition of Matrigel improves the growing environment of tumor spheroids, making it closer to that of in vivo tumor conditions.
Literature	GENERIF_SUMMARY	MAT2B was markedly increased in osteosarcoma (OS) specimens compared with the normal bone tissues, and it was additionally abundantly expressed in OS cell lines. Inhibition of MAT2B expression caused a marked decrease in proliferation and significant increase in apoptosis. Results demonstrated that MAT2B functions as a tumor oncogene in OS in vivo and in vitro via regulation of EGFR and PCNA.
Literature	GENERIF_SUMMARY	S phase-specific degradation of p21 by PCNA-CRL4(cdt2) is sufficient to explain these heterogeneous responses to DNA damage and cell cycle regulation in individual cells.
Literature	GENERIF_SUMMARY	The expressions of CD24, B7-H4 and PCNA have correlations with the occurrence, development, invasion and metastasis of ovarian cancer, and the combined detection may have clinical guiding significance for early diagnosis of ovarian cancer and screening of high-risk patients.
Literature	GENERIF_SUMMARY	Our systematic multidisciplinary approach expands current views on PCNA interactions and reveals that the PIP-box affinity can be modulated over four orders of

Main Category	Category	Terms
		magnitude by positive charges in the flanking regions.
Literature	GENERIF_SUMMARY	These data provide structural insight into the recruitment of DNMT1 to replication sites by PCNA.
Literature	GENERIF_SUMMARY	in Basaloid squamous cell carcinomas , SOX2 and PCNA had diffuse staining patterns in almost all cancer cells. SOX2 overexpression had little effect on the expression levels of claudin4, occludin and PCNA.
Literature	GENERIF_SUMMARY	Expression of PCNA, Ki-67 and COX-2 in breast cancer based on DCE-MRI image information.
Literature	GENERIF_SUMMARY	Long non-coding RNA AK027294 promotes tumor growth by upregulating PCNA in gastric cancer.
Literature	GENERIF_SUMMARY	Data showed that proliferating cell nuclear antigen (PCNA) associated with neutrophil cytosolic factor 1 (p47phox), a key subunit of NADPH oxidase in neutrophils and that this association regulated reactive oxygen species (ROS) production.
Literature	GENERIF_SUMMARY	Changes in Protein Glycosylation as a Result of Aptamer Interactions with Cancer Cells.
Literature	GENERIF_SUMMARY	GSK3beta interacted with PCNA in H1299 lung adenocarcinoma cells. Overexpression of PCNA protein decreased GSK3beta Ser9 phosphorylation. Downregulation of PCNA sensitized H1299 cells to LiCl-induced apoptosis.
Literature	GENERIF_SUMMARY	IL8 can regulate the radiation resistance of ESCC cells and expression of PCNA
Literature	GENERIF_SUMMARY	ATAD5, a PCNA unloader, plays multiple functions at stalled forks including promoting its restart. ATAD5 depletion increases genomic instability upon hydroxyurea treatment in cultured cells and mice. ATAD5 recruits RAD51 to stalled forks in an ATR kinase-dependent manner by hydroxyurea-enhanced protein-protein interactions and timely removes PCNA from stalled forks for RAD51 recruitment.
Literature	GENERIF_SUMMARY	PCNA Unloading Is Negatively Regulated by BET Proteins.
Literature	GENERIF_SUMMARY	GTSE1, together with CDC20, PCNA, and MCM6, may synergistically promote adverse prognosis in hepatocellular carcinoma by activating cell cycle.
Literature	GENERIF_SUMMARY	Human PCNA Structure, Function and Interactions.
Literature	GENERIF_SUMMARY	high protein and mRNA levels closely related to occurrence, development, and prognosis non-small cell lung cancer
Literature	GENERIF_SUMMARY	Identification of PCNA-interacting protein motifs in human DNA polymerase delta.
Literature	GENERIF_SUMMARY	CDK-Independent and PCNA-Dependent Functions of p21 in DNA Replication.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Histone deacetylase 1 regulates the malignancy of oral cancer cells via miR-154-5p/PCNA axis.
Literature	GENERIF_SUMMARY	Nuclear IGF1R interact with PCNA to preserve DNA replication after DNA-damage in a variety of human cancers.
Literature	GENERIF_SUMMARY	Succinylation at a key residue of FEN1 is involved in the DNA damage response to maintain genome stability.
Literature	GENERIF_SUMMARY	PCNA activates the MutLgamma endonuclease to promote meiotic crossing over.
Literature	GENERIF_SUMMARY	TRIM28 functions as the SUMO E3 ligase for PCNA in prevention of transcription induced DNA breaks.
Literature	GENERIF_SUMMARY	Histone H3K79 demethylation by KDM2B facilitates proper DNA replication through PCNA dissociation from chromatin.
Literature	GENERIF_SUMMARY	Dynamic DNA-bound PCNA complexes co-ordinate Okazaki fragment synthesis, processing and ligation.
Literature	GENERIF_SUMMARY	PCNA Monoubiquitination Is Regulated by Diffusion of Rad6/Rad18 Complexes along RPA Filaments.
Literature	GENERIF_SUMMARY	Role of LLT1 and PCNA as Natural Killer Cell Immune Evasion Strategies of HCT 116 Cells.
Literature	GENERIF_SUMMARY	miR-1301-3p suppresses tumor growth by downregulating PCNA in thyroid papillary cancer.
Literature	GENERIF_SUMMARY	Toxoplasma gondii Extends the Life Span of Infected Human Neutrophils by Inducing Cytosolic PCNA and Blocking Activation of Apoptotic Caspases.
Literature	GENERIF_SUMMARY	PCNA, a focus on replication stress and the alternative lengthening of telomeres pathway.
Literature	GENERIF_SUMMARY	Expressions of CD44, PCNA and MRP1 in lung cancer tissues and their effects on proliferation and invasion abilities of lung cancer cell line 95D.
Literature	GENERIF_SUMMARY	Proliferating cell nuclear antigen directly interacts with androgen receptor and enhances androgen receptormediated signaling.
Literature	GENERIF_SUMMARY	Unlocking the PIP-box
Literature	GENERIF_SUMMARY	Expressions of cytokeratin 14 and proliferating cell nuclear antigen in the Hertwig's epithelial root sheath of a Vps4b knockout mouse.
Literature	GENERIF_SUMMARY	Transcription-coupled structural dynamics of topologically associating domains regulate replication origin efficiency.
Literature	GENERIF_SUMMARY	PCNA inhibition enhances the cytotoxicity of beta-lapachone in NQO1-Positive cancer cells by augmentation of oxidative stress-induced DNA damage.
Literature	GENERIF_SUMMARY	Cytoskeleton integrity influences XRCC1 and PCNA

Main Category	Category	Terms
		dynamics at DNA damage.
Literature	GENERIF_SUMMARY	Evading immune surveillance via tyrosine phosphorylation of nuclear PCNA.
Literature	GENERIF_SUMMARY	Cryo-EM structure of human Pol kappa bound to DNA and mono-ubiquitylated PCNA.
Literature	GENERIF_SUMMARY	Post-Translational Modifications of PCNA in Control of DNA Synthesis and DNA Damage Tolerance-the Implications in Carcinogenesis.
Literature	GENERIF_SUMMARY	Ring finger protein 6 enhances chemo-resistance by transcriptionally activating proliferating cell nuclear antigen expression and attenuating DNA damage in lung adenocarcinoma.
Literature	GENERIF_SUMMARY	Can PCNA and LIM15 gene expression levels predict sperm retrieval success in men with non-obstructive azoospermia?
Literature	GENERIF_SUMMARY	Characterization of the Interaction Between SARS-CoV-2 Membrane Protein (M) and Proliferating Cell Nuclear Antigen (PCNA) as a Potential Therapeutic Target.
Literature	GENERIF_SUMMARY	Prognostic value of p16, p53, and pcna in sarcoma and an evaluation of immune infiltration.
Literature	GENERIF_SUMMARY	Ubiquitinated PCNA Drives USP1 Synthetic Lethality in Cancer.
Literature	GENERIF_SUMMARY	DUT enhances drug resistance to proteasome inhibitors via promoting mitochondrial function in multiple myeloma.
Literature	GENERIF_SUMMARY	PCNA regulates primary metabolism by scaffolding metabolic enzymes.
Literature	GENERIF_SUMMARY	HNF4G stimulates the development of pancreatic cancer by promoting IGF2BP2 transcription.
Literature	GENERIF_SUMMARY	Effects of Defective Unloading and Recycling of PCNA Revealed by the Analysis of ELG1 Mutants.
Literature	GENERIF_SUMMARY	C/EBPbeta expression decreases in cervical cancer and leads to tumorigenesis.
Literature	GENERIF_SUMMARY	Unchanged PCNA and DNMT1 dynamics during replication in DNA ligase I-deficient cells but abnormal chromatin levels of non-replicative histone H1.
Literature	GENERIF_SUMMARY	Molecular insight into the PCNA-binding mode of FBH1.
Literature	GENERIF_SUMMARY	A thermosensitive PCNA allele underlies an ataxia-telangiectasia-like disorder.
Literature	GENERIF_SUMMARY	Ubiquitin specific peptidase 37 and PCNA interaction promotes osteosarcoma pathogenesis by modulating replication fork progression.
Literature	GENERIF_SUMMARY	Short-range end resection requires ATAD5-mediated

Main Category	Category	Terms
		PCNA unloading for faithful homologous recombination.
Literature	GENERIF_SUMMARY	G-CSF reshapes the cytosolic PCNA scaffold and modulates glycolysis in neutrophils.
Literature	GENERIF_SUMMARY	FANCD2-dependent mitotic DNA synthesis relies on PCNA K164 ubiquitination.
Literature	GENERIF_SUMMARY	Checkpoint activation by Spd1
Literature	GENERIF_SUMMARY	Squalene monooxygenase facilitates bladder cancer development in part by regulating PCNA.
Literature	GENERIF_SUMMARY	Loci on chromosome 20 interact with rs16969968 to influence cigarettes per day in European ancestry individuals.
Literature	GENERIF_SUMMARY	Expression of CD44, PCNA and E-cadherin in pterygium tissues.
Literature	GENERIF_SUMMARY	Cryo-EM reveals a nearly complete PCNA loading process and unique features of the human alternative clamp loader CTF18-RFC.
Literature	GENERIF_SUMMARY	EIF3B stabilizes PCNA by counteracting SYVN1-mediated ubiquitination to serve as a promotor in cholangiocarcinoma.
Literature	GENERIF_SUMMARY	DNAJC24 acts directly with PCNA and promotes malignant progression of LUAD by activating phosphorylation of AKT.
Literature	GENERIF_SUMMARY	Polyubiquitinated PCNA triggers SLX4-mediated break-induced replication in alternative lengthening of telomeres (ALT) cancer cells.
Pathways	KEGG_PATHWAY	DNA replication
Pathways	KEGG_PATHWAY	Base excision repair
Pathways	KEGG_PATHWAY	Nucleotide excision repair
Pathways	KEGG_PATHWAY	Mismatch repair
Pathways	KEGG_PATHWAY	Cell cycle
Pathways	KEGG_PATHWAY	Tight junction
Pathways	KEGG_PATHWAY	Hepatitis B
Pathways	REACTOME_PATHWAY	Translesion synthesis by REV1
Pathways	REACTOME_PATHWAY	Translesion synthesis by Y family DNA polymerases bypasses lesions on DNA template
Pathways	REACTOME_PATHWAY	Recognition of DNA damage by PCNA-containing replication complex
Pathways	REACTOME_PATHWAY	Translesion Synthesis by POLH
Pathways	REACTOME_PATHWAY	Resolution of AP sites via the multiple-nucleotide patch replacement pathway

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Transcription of E2F targets under negative control by DREAM complex
Pathways	REACTOME_PATHWAY	G0 and Early G1
Pathways	REACTOME_PATHWAY	Telomere Maintenance
Pathways	REACTOME_PATHWAY	Cell Cycle
Pathways	REACTOME_PATHWAY	Polymerase switching on the C-strand of the telomere
Pathways	REACTOME_PATHWAY	Processive synthesis on the C-strand of the telomere
Pathways	REACTOME_PATHWAY	Telomere C-strand (Lagging Strand) Synthesis
Pathways	REACTOME_PATHWAY	Removal of the Flap Intermediate from the C-strand
Pathways	REACTOME_PATHWAY	Extension of Telomeres
Pathways	REACTOME_PATHWAY	Generic Transcription Pathway
Pathways	REACTOME_PATHWAY	SUMOylation
Pathways	REACTOME_PATHWAY	SUMO E3 ligases SUMOylate target proteins
Pathways	REACTOME_PATHWAY	Transcriptional Regulation by TP53
Pathways	REACTOME_PATHWAY	Metabolism of proteins
Pathways	REACTOME_PATHWAY	Mitotic G1 phase and G1/S transition
Pathways	REACTOME_PATHWAY	SUMOylation of DNA replication proteins
Pathways	REACTOME_PATHWAY	Mismatch Repair
Pathways	REACTOME_PATHWAY	Mismatch repair (MMR) directed by MSH2:MSH6 (MutSalpha)
Pathways	REACTOME_PATHWAY	Mismatch repair (MMR) directed by MSH2:MSH3 (MutSbeta)
Pathways	REACTOME_PATHWAY	PCNA-Dependent Long Patch Base Excision Repair
Pathways	REACTOME_PATHWAY	Translesion synthesis by POLK
Pathways	REACTOME_PATHWAY	Translesion synthesis by POLI
Pathways	REACTOME_PATHWAY	Termination of translesion DNA synthesis
Pathways	REACTOME_PATHWAY	HDR through Homologous Recombination (HRR)
Pathways	REACTOME_PATHWAY	DNA Double-Strand Break Repair
Pathways	REACTOME_PATHWAY	Homology Directed Repair
Pathways	REACTOME_PATHWAY	HDR through Homologous Recombination (HRR) or Single Strand Annealing (SSA)
Pathways	REACTOME_PATHWAY	Gap-filling DNA repair synthesis and ligation in GG-NER
Pathways	REACTOME_PATHWAY	Nucleotide Excision Repair
Pathways	REACTOME_PATHWAY	Global Genome Nucleotide Excision Repair (GG-NER)

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Dual Incision in GG-NER
Pathways	REACTOME_PATHWAY	Post-translational protein modification
Pathways	REACTOME_PATHWAY	Transcription-Coupled Nucleotide Excision Repair (TC-NER)
Pathways	REACTOME_PATHWAY	Dual incision in TC-NER
Pathways	REACTOME_PATHWAY	Gap-filling DNA repair synthesis and ligation in TC-NER
Pathways	REACTOME_PATHWAY	TP53 Regulates Transcription of Cell Cycle Genes
Pathways	REACTOME_PATHWAY	TP53 Regulates Transcription of Genes Involved in G2 Cell Cycle Arrest
Pathways	REACTOME_PATHWAY	Polymerase switching
Pathways	REACTOME_PATHWAY	Leading Strand Synthesis
Pathways	REACTOME_PATHWAY	Removal of the Flap Intermediate
Pathways	REACTOME_PATHWAY	Processive synthesis on the lagging strand
Pathways	REACTOME_PATHWAY	Lagging Strand Synthesis
Pathways	REACTOME_PATHWAY	DNA strand elongation
Pathways	REACTOME_PATHWAY	G1/S-Specific Transcription
Pathways	REACTOME_PATHWAY	G1/S Transition
Pathways	REACTOME_PATHWAY	Synthesis of DNA
Pathways	REACTOME_PATHWAY	S Phase
Pathways	REACTOME_PATHWAY	Cell Cycle, Mitotic
Pathways	REACTOME_PATHWAY	DNA Replication
Pathways	REACTOME_PATHWAY	RNA Polymerase II Transcription
Pathways	REACTOME_PATHWAY	Base Excision Repair
Pathways	REACTOME_PATHWAY	Chromosome Maintenance
Pathways	REACTOME_PATHWAY	DNA Damage Bypass
Pathways	REACTOME_PATHWAY	DNA Repair
Pathways	REACTOME_PATHWAY	Resolution of Abasic Sites (AP sites)
Pathways	REACTOME_PATHWAY	Gene expression (Transcription)
Pathways	REACTOME_PATHWAY	Protein ubiquitination
Pathways	REACTOME_PATHWAY	E3 ubiquitin ligases ubiquitinate target proteins
Pathways	WIKIPATHWAYS	Cell cycle
Pathways	WIKIPATHWAYS	Retinoblastoma gene in cancer
Pathways	WIKIPATHWAYS	Primary focal segmental glomerulosclerosis FSGS

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Nifedipine activity
Pathways	WIKIPATHWAYS	Oxidative damage response
Pathways	WIKIPATHWAYS	DNA IR double strand breaks and cellular response via ATM
Pathways	WIKIPATHWAYS	DNA IR damage and cellular response via ATR
Pathways	WIKIPATHWAYS	EGF EGFR signaling
Pathways	WIKIPATHWAYS	G1 to S cell cycle control
Pathways	WIKIPATHWAYS	DNA replication
Pathways	WIKIPATHWAYS	Hepatitis B infection
Pathways	WIKIPATHWAYS	Base excision repair
Pathways	WIKIPATHWAYS	Nucleotide excision repair
Pathways	WIKIPATHWAYS	7q11 23 copy number variation syndrome
Pathways	WIKIPATHWAYS	DNA repair pathways full network
Pathways	WIKIPATHWAYS	16p11 2 proximal deletion syndrome
Pathways	WIKIPATHWAYS	p53 transcriptional gene network
Pathways	WIKIPATHWAYS	Nucleotide excision repair in xeroderma pigmentosum
Pathways	WIKIPATHWAYS	Urotensin II mediated signaling
Pathways	WIKIPATHWAYS	DNA mismatch repair
Pathways	WIKIPATHWAYS	10q11 21q11 23 copy number variation syndrome
Pathways	WIKIPATHWAYS	Macrophage stimulating protein MSP signaling
Pathways	WIKIPATHWAYS	NF1 copy number variation syndrome
Pathways	WIKIPATHWAYS	Rubinstein Taybi syndrome 1
Pathways	WIKIPATHWAYS	Nucleosome remodeling complex
Pathways	WIKIPATHWAYS	Senescence and autophagy in cancer
Protein Domains	INTERPRO	Pr_cel_nuc_antig
Protein Domains	INTERPRO	Pr_cel_nuc_antig_N
Protein Domains	INTERPRO	Pr_cel_nuc_antig_C
Protein Domains	INTERPRO	Pr_cel_nuc_antig_CS
Protein Domains	INTERPRO	DNA_clamp_sf